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A Multi-Component Deep Learning Approach for Cyber-Attack Detection in Critical Infrastructure SCADA Systems

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Abstract

Industrial control networks based on Supervisory Control and Data Acquisition (SCADA) systems used in critical infrastructure systems require reliable, scalable, and generalizable attack detection mechanisms in the face of increasing cyber-threats. However, a large portion of the datasets commonly used in the literature are based on general-purpose network traffic and do not adequately reflect the cyclical, deterministic, and process-oriented communication structure of industrial protocols such as Modbus/TCP. In this study, a completely isolated virtual SCADA environment was designed to address this limitation. A SCADA-specific multi-class attack dataset called SCADANet was developed, containing 13 different attack scenarios generated concurrently with legitimate network traffic. In this context, a hybrid deep learning architecture consisting of parallel dilated Convolutional Neural Network (CNN) structures, Residual-SE blocks, and Long Short-Term Memory (LSTM) layers, capable of modeling multi-scale spatial patterns and feature-level sequential and contextual relationships within transformed traffic representations, was proposed. To examine the generalizability of the proposed approach, the developed model was tested on both the SCADANet dataset and the WUSTL-IIoT-2021 dataset, which is widely used in the literature. Experimental results show that the proposed architecture achieves high accuracy and macro-F1-score on both datasets, particularly demonstrating a significant increase in sensitivity in minority attack classes. Furthermore, the ablation study conducted reveals that each component in the architecture contributes meaningfully and complementarily to the overall performance. The findings demonstrate that the proposed deep learning-based approach can effectively model traffic characteristics specific to SCADA systems and exhibits stable attack detection performance across different data sources. In this respect, the study makes important methodological and experimental contributions to the literature on cyber-attack detection in SCADA-based critical infrastructure systems.

Keywords: SCADA security; critical infrastructure systems; hybrid deep learning; cyber-attack detection

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1. Introduction

Systems with critical infrastructure such as energy, healthcare, transportation, and financial services are essential and necessary for the natural flow of life (see Figure 1). A large portion of such systems are managed using Supervisory Control and Data Acquisition (SCADA) and Industrial Control System (ICS) technologies [1,2]. The ability to manage systems remotely creates a natural cybersecurity requirement within the system. To ensure

the continuity of the services provided by critical infrastructure, the system must be protected against external attacks [3].

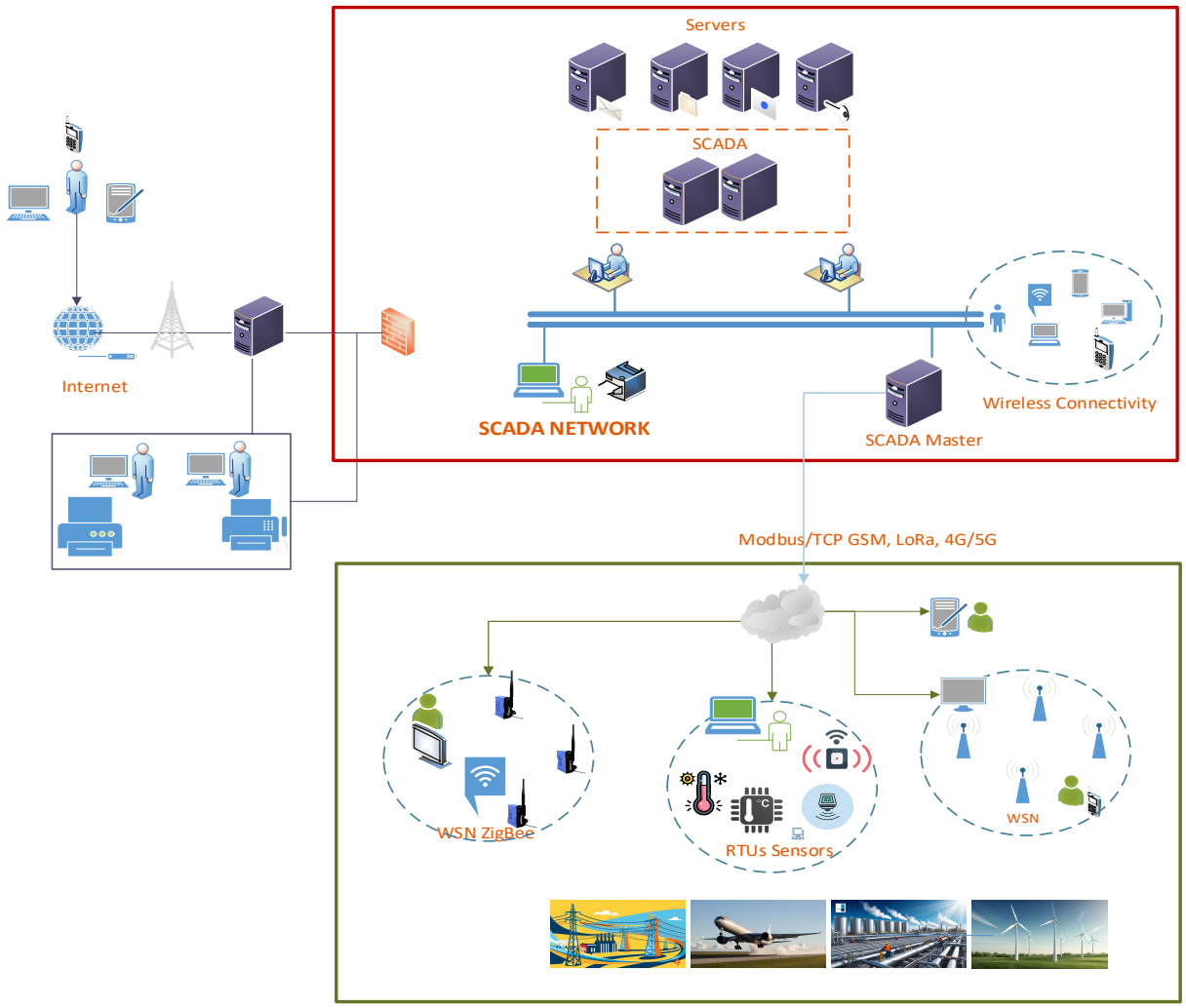


Figure 1. A general SCADA system architecture illustrating open communication channels and interconnected industrial devices that could be targets of cyberattacks.

A vulnerability in the system does not only cause a malfunction in the system. External factors connected to the system may also suffer damage [4]. At the forefront of these threats are Denial of Service (DoS) attacks and Advanced Persistent Threats (APTs). These attacks occur frequently, slowing down or obstructing the operation of the system [5,6].

Attack and anomaly detection in SCADA-based systems remains a critical issue due to real-time operational requirements, low tolerance for false positives, and uneven traffic distributions. Furthermore, the multidimensional and heterogeneous nature of industrial network traffic, combined with attacks hidden within legitimate communication flows, makes it difficult to reliably detect attacks under realistic operational conditions [7–10].

There is a growing need for real-time intrusion detection systems with low false alarm rates in Industrial Control Systems (ICSs), particularly SCADA networks. Traditional security systems are inadequate against evolving attacks. Attackers embedding attacks within normal traffic flows and class imbalance in datasets further increase the need to research more effective methods. Advanced methods such as deep learning ensure not only high accuracy but also high sensitivity in classes with data imbalance. Therefore, development of new and effective deep learning models will be highly valuable for protecting such systems. This study proposes a SCADANet dataset that realistically represents different

attack scenarios specific to SCADA. Furthermore, a hybrid deep learning architecture capable of modelling multi-scale and time dependencies well in attacks is proposed.

This study differs from these approaches by creating a SCADA-specific dataset in an isolated virtual environment and evaluating the proposed model on both SCADANet and WUSTL-IIoT-2021 datasets. The proposed architecture integrates Parallel Dilated Conv1D, Residual-SE blocks, and LSTM layers, enabling it to represent multi-scale and feature-level contextual traffic.

The contributions of this study can be summarised as follows:

- ***SCADA-specific SCADANet dataset:*** A virtual SCADA environment was developed in a completely isolated setting with a SCADA-based structure. In this environment, Modbus/TCP and ICMP/HTTP, which are present in background traffic, were generated together. Attack scenarios were generated concurrently and embedded within the normal network traffic flow.
- ***Flow richness:*** Multi-layered attributes were extracted for raw packets on the network. JA4+-based fingerprint features capable of distinguishing attacks with particularly low volume in network traffic were obtained.
- ***Novel hybrid architecture:*** Parallel dilated Conv1D was used to extract short- and long-range features in the SCADA system. A Residual-SE block was added to increase representational power. A hybrid method that also learns time dependencies using LSTM was presented.
- ***A balanced approach to imbalanced data classes:*** Despite data imbalance, a two-stage optimisation was applied using Focal Loss and class weights. This has ensured increased sensitivity in minority classes.
- ***Cross-dataset generalizability:*** The proposed hybrid method was applied both to the created dataset and to the WUSTL-IIoT-2021 dataset [11], which is frequently used in the literature. This demonstrates that the proposed hybrid method is not dependent on the dataset and exhibits strong generalization capabilities.

Section 2 reviews relevant studies on cyberattack detection and current research gaps. Section 3 describes the experimental study processes. This section presents the data preprocessing steps and details of the proposed deep learning architecture. Section 4 presents the experimental setup, dataset creation process, evaluation metrics, results, ablation analysis, and model interpretability analysis. Section 5 compares the proposed structure with recent studies in the literature. Section 6 concludes the article and addresses the limitations of the study. Future research is also discussed.

2. Background and Literature Review

2.1. Cyber-Attacks Targeting SCADA and Industrial Control Systems

Cyberattacks targeting Industrial Control Systems (ICSs), particularly SCADA systems, are increasing day by day. These attacks target different points within such systems. Each type of attack has its own target. Reconnaissance attacks gather information about the SCADA system and map its architectural structure. Weak points in the architecture are identified by scanning with network tools [12]. In command and measurement injection attacks, commands are sent to manipulate the system. These deceptive commands cause adverse behaviour in the system and can lead to physical damage and system interruptions. DoS attacks flood the system with unnecessary data traffic. This causes disruption to the services running on the system. They aim to render the system unusable due to protocol weaknesses [13]. Malicious software attacks attempt to gain unauthorised access. Once access is gained, data is stolen or used to interrupt operations [14]. Hijacking and blackout attacks focus on taking control of the system and shutting it down completely. Traditional SCADA and ICS environments often rely on legacy industrial communication

protocols that lack built-in security mechanisms, making them vulnerable to protocol-based attacks [15]. Replay and fragmentation attacks focus on capturing data packets. These packets are resent, causing process interruptions [7,16]. Such attacks have caused significant operational and economic damage worldwide. Cybersecurity is critically important for critical infrastructures, particularly Industrial Control Systems (ICSs) such as SCADA systems. Detecting attacks on this infrastructure is more complex and difficult than detecting attacks on traditional environments [17]. Industrial systems lack inherent security features such as Modbus/TCP, encryption, authentication, and integrity checks. This makes such systems vulnerable to various attacks [18]. ARP packets play a critical role in maintaining communication between devices in critical network infrastructures such as SCADA systems. A review of the literature indicates that ARP spoofing-based man-in-the-middle attacks can enable attackers to monitor network traffic, manipulate packets, and inject unauthorized data [19,20]. Advanced attack techniques such as Flood Attacks, Wrong Command Injection, and DoS attacks pose serious threats to industrial systems such as SCADA. In flood attacks, attackers corrupt Modbus/TCP packets to prevent the controller from interpreting commands or to cause unsafe scenarios [18]. In false command injection attacks, attackers attempt to inject commands by hiding behind repetitive responses from the SCADA operator. This can cause significant damage [8]. While traditional DoS attacks focus on disrupting the service on the system, protocol-specific DoS attacks focus on security vulnerabilities in Modbus/TCP protocols in order to disrupt communication [9]. Some of these attacks are summarised in Table 1.

Table 1. Some attacks on SCADA and ICS systems and their effects.

Name of Attack	Target	Attack Centre	Year	Country/Region	Impact
Stuxnet [21]	SCADA/ICS	Iran's Natanz Nuclear Facility	2010	Iran	Physical damage and disruption of operations in centrifuges through manipulation of PLC/SCADA systems
BlackEnergy (2015 Ukraine Power Grid) [22]	SCADA/ICS	Ukrainian Electricity Grid (transmission/distribution)	2015	Ukraine	Power outages (affecting approximately 230,000 users) caused the remote SCADA system to shut down.
Triton/Trisis [23]	SCADA/ICS	Triconex SIS in a petrochemical plant in the Middle East	2017	Middle East (Arabia)	Security was disabled by targeting the SIS logic.
Havex (Backdoor.Oldrea) [24]	SCADA/ICS	European energy companies	2013–2014	Europe (primarily the energy sector)	Information gathered for reconnaissance—sabotage in ICS/OPC environments
Dragonfly (Energetic Bear) [25]	SCADA/ICS	European & US energy companies	2010	Global (critical infrastructure/energy)	Supply chain + spearphishing + drive-by to infiltrate and explore the system
IDS Evasion via Fragroute [26]	IDS	General IDS systems	1998/2002	—	Misinterpretation of traffic by IDS due to packet fragmentation/rewriting, etc., detection prevention
Snort IDS Evasion [26]	IDS	Networks using Snort	1998+	—	Due to differences such as reassembly/fragmentation, Snort risks missing some scenarios (if there is no correct pre-processor/normaliser).
Shamoon [27]	SCADA (indirect)	Saudi Aramco (IT network; OT integrations may be affected)	2012	Saudi Arabia	Widespread wiper effect; reports indicate that approximately 30,000 computers wer affected.

Attacks on SCADA and ICS systems, which cause physical and operational damage, are detailed in Table 1. The extent of the damage these attacks cause to countries and people is evident.

2.2. Traditional SCADA Attack Detection Methods

Cyberattacks targeting Industrial Control Systems, including SCADA systems, cause significant physical and operational damage. Various methods and approaches have been developed to prevent these attacks. Efforts to prevent and detect attacks on systems are documented in the literature as traditional detection methods, machine learning methods, deep learning-based methods, and hybrid methods. Among these methods, traditional attack detection includes signature-based intrusion detection systems (IDSs), rule-based approaches, and anomaly-based statistical models.

IDS-based systems are designed to prevent unauthorised access based on predefined patterns and signatures of known threats. It is an effective method for detecting known attacks. However, it is not successful in detecting new or unknown threats. They are preferred because they are simple and have a low false-positive rate [28]. A study by Tripathi and colleagues indicated that zero-day attacks and new types of unauthorised access techniques, which are becoming widespread in SCADA systems, cannot be detected by traditional signature-based detection systems [13,29,30]. In rule-based detection systems, experts or past information are used to identify suspicious activities. In his study, Sayghe indicated that rule-based systems can be successful in certain operations [31]. Okpeseyi and colleagues, however, stated that rule-based systems are weak against evolving threats and that rules in such systems need to be constantly updated [16,32].

In statistical detection methods based on anomalies, deviations from normal behaviour patterns are taken into account. Statistical models and deviations are evaluated to define abnormal behaviour. Known attacks are not trusted for general purposes. Therefore, it provides better detection of new attacks compared to other traditional models. This approach appears to be slightly more suitable for dynamic environments such as SCADA [33]. Ahakonye and colleagues have stated that detection using statistical models in SCADA systems is a challenging process, as natural flows can be mistakenly perceived as threats. For this reason, they have demonstrated a tendency to produce false positives [7,34,35].

Traditional detection systems show that while it is possible to detect known attacks in the system, it is difficult to detect new types of attacks using such methods.

2.3. Machine Learning-Based SCADA Attack Detection

Unlike traditional methods, machine learning methods are more successful in detecting attacks on SCADA systems. Various machine learning methods offer advantages. The literature includes machine learning methods that enable the detection of attacks on SCADA systems, such as Support Vector Machines (SVMs), Decision Trees (DTs), Random Forest, k-Nearest Neighbour (k-NN), and XGBoost. In their study, Oyucu and colleagues demonstrated that ensemble learning-based approaches can effectively detect volumetric attacks, such as DDoS, in SDN-based SCADA systems with high classification performance [36]. Sukhadeo et al. classified attacks using random forest and k-KNN on the NSL-KDD dataset [37]. Lee et al. achieved a result of 95% in detecting attacks in SCADA systems using SVM and XGBoost methods [38]. When the studies are examined, it is seen that feature extraction has a high impact on performance in machine learning-based detection. In another study employing various feature extractions, redundant fields were eliminated and generalizability was improved [38]. Overall, the achieved performance exceeds 90% in the studies conducted. Some examples of studies using machine learning methods are shown in Table 2.

Studies conducted on machine learning methods have shown that they achieve higher performance than traditional methods. Research indicates that the most significant reason for this performance is feature selection.

Table 2. Studies conducted using various machine learning methods and feature selections.

Method	Feature Selection	Rate of Success	Number of Classes	Description
SVM [38]	PCA	%93.5	5	Linear SVM worked with high accuracy for normal and 4 attack classes in SCADA data
Neural Network/LightGBM [39]	Genetic Algorithm	%98.60	5	GA-based feature selection reduces the number of redundant features in ICS/SCADA datasets. It also improves the performance of unauthorised access detection.
Random Forest [40]	Information Gain	%95.1	4	High performance with low overfitting on SCADA network traffic
Random Forest [37]	Chi-Square	%97.3	5	NSL-KDD-based IDS design; robust against class imbalance
k-NN [38]	Correlation Based FS	%92.4	4	Based on PLC data; the best results were obtained with k = 5
k-NN [41]	Relief-F	%94.2	3	More stable results with weighted distance metrics in the SCADA system
XGBoost [42]	PCA + IQR	%98.5	9	Thanks to pre-processing based on PCA and IQR, anomaly detection performance against various types of cyber-attacks in DNP3 SCADA attack data has been improved.
LightGBM [43]	L1 Regularization	%95.4	5	Similar accuracy with lower resource usage compared to XGBoost

2.4. Deep Learning-Based SCADA Security Solutions

More advanced attack detection methods in SCADA-based systems are deep learning-based methods. These include Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM), Auto-Encoder, and Transformer Models. CNN models are widely used for detection in such systems. They may have limitations in capturing long-range sequential relationships or temporal patterns when used alone. However, they stand out due to their computational efficiency and speed [42,44,45]. LSTMs are particularly better at processing sequence-structured representations. This provides more effective learning in detecting related attacks. They require strong recall rates and computational resources [43,46–48]. Autoencoder-based methods have proven to be effective in learning latent representations of abnormal traffic patterns [49]. Transformer models, on the other hand, perform quite well in capturing long-range dependencies and complex sequential patterns [50]. Studies in the existing literature show that deep learning performance depends largely on the characteristics of the dataset and the traffic structure. Each deep learning-based method focuses on specific features within the system. This demonstrates that each model has its own unique structure (see Table 3).

Table 3. Comparison of deep learning-based models used for attack detection in SCADA-based systems.

Detection Accuracy	Efficiency Performance	Temporal–Spatial Feature Extraction	False-Positive/Negative Rates
Transformers demonstrate superior performance to CNNs and RNNs in terms of accuracy and generalisation [49]	Transformer optimised for low-latency inference	Transformers capture long-range dependencies better than RNN/CNN	Low false-positive and false-negative rates
CNN demonstrates superior performance to other DL models across multiple datasets [50]	Suitable efficiency for CNN	CNN is effective for spatial features; attention mechanisms included	False-positive rates considered in the evaluation
1D CNN and RNN demonstrate better performance than traditional ML in anomaly detection [51]	1D CNNs are preferred for long sequences due to their efficiency	1D CNN replaces RNN for spatio-temporal features	Improved false-positive rates compared to traditional methods

Table 3. *Cont.*

Detection Accuracy	Efficiency Performance	Temporal–Spatial Feature Extraction	False-Positive/Negative Rates
The CKAN model achieves over 98% accuracy on the multi-IDS dataset [52]	Fewer parameters increase computational efficiency	Develops CNN feature extraction with block layers	According to CNN, improved false-positive rates
LSTM achieves >99% accuracy; DCNN achieves 96% accuracy on ICS datasets [53]	Improved efficiency with smaller models	LSTM and DCNN are effective for temporal and spatial features	Low false-positive and false-negative rates
SCADA-based unsupervised anomaly detection in gas pipelines [54]	Suitable efficiency for real time	The automatic encoder captures normal traffic characteristics	False-positive and false-negative rates managed by thresholding
Transformer models deliver good performance [55]	Computational efficiency is not discussed.	It excels in transformer series modelling	False-positive and false-negative rates are not detailed
Deep autoencoder detects DDoS attacks with high detection rates and low false alarms [56]	Suitable efficiency for real time	The automatic encoder captures network streaming features	Low false-positive and false-negative rates
LSTM-based IDS detects MiTM and DoS attacks within 2 min [57]	Suitable efficiency for real time	LSTM captures temporal dependencies	Low false-positive and false-negative rates
Artificial intelligence-based anomaly detection on the DNP 3.0 network achieves 99% accuracy [58]	Average efficiency	The automatic encoder captures network traffic characteristics	Low false-positive and false-negative rates
Attention-based variational autoencoder improves anomaly detection on the sWAT dataset [59]	Average efficiency	The attention mechanism develops spatial–temporal feature extraction	False-positive and false negative-rates have been reduced

Each deep learning model reveals certain contextual attributes related to attack detection. This contextuality may vary depending on the model type. Therefore, each model appears to be more suitable for separate detection.

2.5. Hybrid and Multi-Scale Architectural Approaches

Deep learning models better understand the unique context of different attacks. This demonstrates that it is not possible to address all attack types with a single model. Although CNNs are weak in time dependencies, they are highly effective in extracting local and spatial patterns. LSTMs are strong in time dependencies but can be sensitive to noise. Autoencoders are powerful in representation learning but are insufficient for classification on their own. Transformer models are good at long dependencies but require large amounts of data, which increases hardware costs. Deep learning models are weak when used alone. Therefore, higher performance can be achieved by combining different deep learning models to create new models that offer better advantages. Examples of studies conducted with hybrid models are shown in Table 4. These models demonstrate the best performance among existing methods with their achieved performance [60].

Hybrid methods are those in which multiple models are combined to achieve better results, following traditional methods, machine learning-based methods, and deep learning models. They are recognised as one of the most effective approaches for attack detection in Industrial Control Systems (ICSs), particularly SCADA systems.

Table 4. Comparison of hybrid models for attack detection in SCADA-based systems.

Detection Accuracy	Efficiency Performance	Temporal-Spatial Feature Extraction	False-Positive/Negative Rates
F1 and AUC improvements in ICS datasets [61]	Low-efficiency training/inference with transformer integration	Powerful spatio-temporal features with CAE + Transformer	Low false alarm with SVDD optimisation
SCADA ransomware detection ~97% Accuracy, >98% AUC [62]	Middle	Spatio-temporal representations with CNN + LSTM	Low false alarm rate (<1.88%)
Very high Acc/F1 with ConvLSTM [63]	Efficient in practice with a combined structure	ConvLSTM spatial + temporal dependency	Reduced FP rate
99.5% accuracy in DNP3 [64]	Middle	CNN-LSTM	Low FP/FN
In the PLC network, Acc = 95.11, F1 = 95.12 [65]	High productivity cost	Multi-scale CNN + Transformer	FP/FN improved
AE-GRU outperforms independent methods [66]	Middle	GRU-based autoencoder	FP/FN improved
FDI detection improved with CNN-LSTM-AE [67]	Middle	CNN-LSTM + AE	FP has decreased significantly
Semi-supervised LSTM-AE effective [68]	Middle	LSTM Autoencoder	FP/FN improved
GoogleNet-LSTM achieves 97.56% accuracy in Modbus [69]	Middle	CNN-LSTM	Low FP and misses

2.6. Limitations of Existing Studies and Research Gap

Literature studies have shown significant progress in detecting attacks on SCADA systems. Despite these developments, some research gaps have been identified. Based on the findings from these studies, the existing limitations and research gaps can be summarised as follows.

Realistic and SCADA-specific dataset: The literature contains many studies on attack detection in SCADA systems. Studies have used datasets such as NSL-KDD [70] and CIC-IDS2017 [71], which reflect general network traffic. These datasets are based on irregular traffic flows found in normal networks. They do not reflect the cyclical and predictable data flow associated with traffic involving industrial protocols such as Modbus/TCP. In a real SCADA environment, devices are queried at specific intervals. The query–response cycle continues for a long period. The packet size and traffic are generally low. Anomalies usually arise from a command within the packet or a disruption in the timing cycle. Since datasets such as NSL-KDD and CIC-IDS2017 reflect general network traffic, they do not reflect command injection and protocol-based attack vectors. This calls into question the reliability of these datasets as training data for attack detection in SCADA systems.

Failure to Address Imbalanced Data and Diverse Attack Scenarios Together: In SCADA-based systems, normal network traffic is much larger than attack traffic. This leads to significant class imbalance among attack categories. Detecting attack classes that occur infrequently is particularly difficult. This causes the developed model to focus on attacks belonging to the majority class, leading to attacks belonging to the minority class being missed. Comprehensive scenarios for numerous attack classes are limited in existing datasets.

The inadequacy of single models and the need for hybrid architectural models: Deep learning models focus on specific features within a dataset. While a model may perform well on one dataset, another deep learning model may be more effective on a different dataset. CNN models focus on local and spatial patterns, LSTMs on sequential or contextual relationships in structured feature representations, and autoencoders on normal behaviour

representations. An attack occurring in SCADA systems may exhibit both anomalies in the packet internal structure and irregular relationships among traffic-derived attributes or sequence-structured representations. A large portion of existing studies attempt to capture this through a single model. The proposed multi-layer hybrid architecture is designed to detect heterogeneous attack patterns by jointly learning spatial and temporal traffic characteristics.

Insufficient assessment of generalizability and robustness: Literature reviews reveal that studies on detecting attacks in SCADA systems typically focus on a single dataset. This leads to overfitting to the dataset and prevents the same level of success from being achieved in different scenarios and structures. The model proposed in this study was evaluated by applying it to different datasets. The proposed multi-layered architecture was designed to deliver consistent detection performance across heterogeneous SCADA datasets. This study was conducted to address the limitations and gaps mentioned above.

3. Materials and Methods

3.1. Data Preprocessing

The SCADANet dataset used in this study was first subjected to the stages of missing value imputation, feature scaling, and label encoding. Missing values were imputed with a constant value of 0. All numerical features were normalized using StandardScaler. To ensure a reliable evaluation and prevent data leakage between dataset partitions, all preprocessing steps were performed independently within each cross-validation fold. In each fold, the imputer and StandardScaler were trained only on the training data. The learned parameters were then applied to the corresponding validation and test subsets. To prevent unintended information transfer between data partitions, the training and test sets were separated from one another prior to preprocessing. Additionally, by executing preprocessing and feature extraction independently within each layer, the overlap of highly similar data samples across dataset partitions was minimized. This process balanced the impact of variables with different scales on the model. It also ensured a reliable model evaluation. The normalized input data was then converted to the (number of examples, number of features, 1) format to make it compatible with the CNN-based architecture. Finally, the output labels were encoded using a LabelEncoder.

Additionally, the data was converted into a one-hot-encoded representation suitable for multi-class classification. To assign greater weight to minority classes during training, the class_weight approach was used to mitigate the impact of class imbalance.

3.2. Proposed Model Architecture

In this study, a hybrid deep learning architecture consisting of Parallel Dilated Conv1D, Residual-SE blocks, and LSTM layers is proposed to jointly model multi-scale spatial patterns and sequential contextual relationships among traffic-derived features in transformed SCADA representations. The general flow diagram of the proposed architecture is presented in Figure 2.

Input Layer: The model's input is organised so that it is of dimension (T, 1) for each example. Here, T represents the time step or feature length.

By clarifying the relationship between the extracted SCADA traffic features and the input representations processed by the multi-layer hybrid architecture, the interpretability of the proposed mathematical formulation is enhanced. Traffic attributes at the protocol level, packet metadata, and JA4+-based fingerprint features are converted into structured feature representations. These features are then processed by parallel dilated convolution and sequential modeling components.

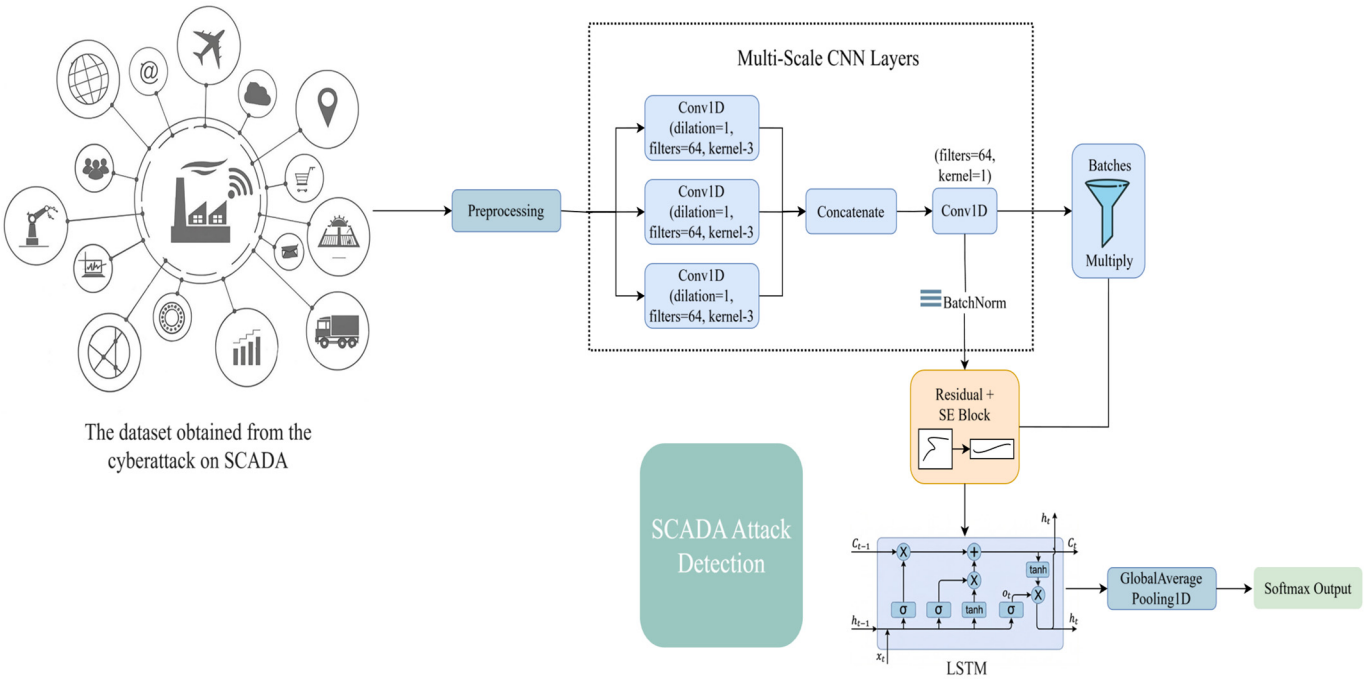


Figure 2. A proposed hybrid multi-layer architecture that combines Parallel Dilated Conv1D, Residual-SE, and LSTM components to capture the multi-scale spatial and temporal relationships of network traffic in critical SCADA systems.

Parallel Dilated Convolution Layers: Convolutional Neural Networks are deep learning architectures frequently used in the field of cyber-security for network traffic analysis, anomaly detection, and DDoS attack classification [72]. CNNs preserve spatial correlations and extract local patterns on the input tensor $X \in \mathbb{R}^{H \times W \times C}$. A convolution operation is defined using a kernel filter $k \in \mathbb{R}^{n \times n \times C}$ as shown in Equation (1).

$$Y(i,j) = \sum_{m=1}^{n} \sum_{n=1}^{n} X(i+m, j+n).k(m,n) \quad (1)$$

In Equation (1), $Y(i,j)$ represents the feature map, $X(i+m, j+n)$ represents the local pixels in the input tensor, and $k(m,n)$ represents the learnable weights multiplied by these pixels. This classical structure can only learn short-range dependencies due to its limited receptive field. In high-dimensional network traffic flows, this makes it difficult to capture long-range dependencies. To overcome this limitation, the dilated convolution mechanism is used. In this approach, gaps are added between the kernel elements with the dilation ratio d. Thus, the effective receptive field is expanded while the number of parameters is kept constant. A one-dimensional dilated convolution operation is mathematically expressed as shown in Equation (2).

$$\left(F^{(d)} * k\right) t = \sum_{\tau} k_{\tau} \cdot F_{t-d\tau} \quad (2)$$

In Equation (2), F represents the input signal (traffic flow signal, etc.), k represents the core weight, d represents the dilation ratio, and τ represents the core's position index.

In this case, the effective receptive field R_{eff} is expressed as in Equation (3).

$$R_{\text{eff}} = [(n-1)d + 1]^2 \quad (3)$$

This increase enables the model to capture broader contextual information with the same number of parameters. This makes it possible to learn both local (fine-grained) and global features simultaneously [73].

In the Efficient Spatial Pyramid module, multi-scale feature extraction is performed by running convolution branches with different dilation rates in parallel. If the dilation rate is defined as $d_k = 2^{k-1}$, the effective receptive field expands as shown in Equation (4).

$$R_{ESP} = \left[(n-1)2^{K-1} + 1\right]^2 \tag{4}$$

The number of parameters for the Efficient Spatial Pyramid module is expressed as in Equation (5).

$$P_{ESP} = \frac{1}{K}\left(NM + n^2N^2\right) \tag{5}$$

Compared with traditional CNN architectures, the proposed architecture detects broader spatial dependencies with a lower computational load.

The generalised form of this structure is referred to as the Parallel Dilated Convolutional Neural Network architecture. This architecture simultaneously runs multiple convolutional layers with different dilation rates (d_i). The feature extraction process for each dilation rate is expressed as in Equation (6).

$$D_c^{(i)} = \sigma\left(F^{d_i} * X\right) \tag{6}$$

Equation (6) denotes σ, which represents non-linear activation functions such as ReLU or Swish. The resulting multi-scale feature maps are then combined in the Global Average Pooling layer. The Global Average Pooling layer is expressed as shown in Equation (7).

$$y_c = \frac{1}{H \times W}\sum_{i=1}^{H}\sum_{j=1}^{W} D_c^{(L)}(i,j) \tag{7}$$

Each step in Equation (7) extracts the average activation of each channel. It also enhances the model's generalisation ability and reduces the risk of overfitting. In cyber-attack detection, low dilation ratios learn local packet statistics and short-range correlations. High dilation rates, on the other hand, capture global anomalies and long-range relationships within network traffic. Thus, the model develops contextual awareness with a wide effective receiver area R_eff. This results in high accuracy and generalisation performance in detecting multi-scale patterns such as DDoS attacks.

During the application phase of the model developed in this study, the input was sent to three parallel Conv1D layers. While the same filter size (3) was used in these layers, different dilation rates were applied. Thus, the architecture was able to learn short-term and long-term dependencies simultaneously. Each branch utilised 64 filters and a ReLU activation function. The outputs obtained from the parallel branches were combined to form a tensor of dimension (T,192). This structure enhanced the model's large-scale feature extraction capacity, thereby strengthening its cyber-attack detection performance [74].

Batch Normalisation and 1 × 1 Convolution: Batch Normalisation (BN) is a widely used normalisation technique to accelerate the training process in deep neural networks while also enhancing its stability. Fundamentally, it aims to reduce internal covariate shift, which arises from the continuous variation in the distribution of activations between layers during training. This method normalises the activations within each mini-batch, transforming them to a mean of zero and a variance of one. It then reshapes them using the scale (γ) and bias (β) parameters that the model can learn [75]. Mathematically, the process is expressed in Equation (8).

$$\mu = \frac{1}{m}\sum_{i=1}^{m} x_i, \qquad \sigma^2 = \frac{1}{m}\sum_{i=1}^{m} (x_i - \mu)^2 \tag{8}$$

Here, the expression $\mu = \frac{1}{m}\sum_{i=1}^{m} x_i$ calculates the mean of the data within the mini-batch. $\sigma^2 = \frac{1}{m}\sum_{i=1}^{m} (x_i - \mu)^2$ calculates the variance of the mini-batch data. The normali-

sation process specified in Equation (9) is then applied. It normalises the data so that the mean is zero and the variance is one.

$$\hat{x_i} = \frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}} \tag{9}$$

The output is obtained by the rescaling and shifting process specified in Equation (10).

$$\mathcal{Y}_i = \gamma \hat{x_i} + \beta, \ \gamma \epsilon \mathbb{R} \ and \ \beta \epsilon \mathbb{R}, \tag{10}$$

Here (Equation (10)), ϵ is a small constant that ensures numerical stability. During the training phase, statistics are calculated over mini-batches, while during the testing phase, the moving average and variance obtained from the entire training set are used. BN enables training with larger learning rates and prevents gradients from exploding or vanishing. It also reduces the risk of overfitting by acting as a regulariser. In the model developed within the scope of this study, the high-dimensional feature map obtained after concatenation was normalised using Batch Normalisation. Subsequently, a 1×1 Conv1D layer (64 filters, ReLU activation) was used to reduce the number of parameters and balance the dimensions [76].

Residual-SE Block: As the number of layers increases in deep neural networks, gradient decay and learning difficulties are observed. The Residual Block, used to overcome this problem, learns the difference between the input and the target instead of the objective function directly [77]. The structure is defined mathematically as shown in Equation (11):

$$y = \mathcal{F}(\mathrm{x}, W) + \mathrm{x} \tag{11}$$

Equation (11) demonstrates the fundamental principle of the residual block. The input x is added to its transformed form $\mathcal{F}(x, W)$ using parameters (W). Thus, the network learns not the target directly, but the difference between the input and the target. The residual block used in this study consists of the following operations:

$$\mathcal{F}(\mathrm{x}) = \mathrm{Conv1D}_{3\times3} \rightarrow \mathrm{Swish} \rightarrow \mathrm{BN} \rightarrow \mathrm{Conv1D}_{1\mathrm{x}1} \tag{12}$$

Equation (12) specifies how the function is calculated. After passing through the 3×3 convolution, Swish activation, and Batch Normalisation steps, a 1×1 convolution is applied. The aim is to transform the input features and obtain more meaningful representations. Shortcut transformation is

$$\mathrm{x}' = \mathrm{Conv1D}_{1\mathrm{x}1}(\mathrm{x}) \tag{13}$$

This expression (Equation (13)) defines the shortcut transformation. The input data is transformed using a 1×1 convolution to achieve dimension matching, thereby enabling it to be added to $\mathcal{F}(x)$.

$$y = \mathrm{Swish}\big(\mathcal{F}(\mathrm{x}) + \mathrm{x}'\big) \tag{14}$$

The Swish activation function performs a smooth non-linear transformation by multiplying the input feature x by the corresponding sigmoid activation output. In this way, rather than completely suppressing moderately negative input values, it maintains the flow of information by producing small but non-zero outputs. Unlike the ReLU activation function, Swish does not completely suppress negative activations or force them to zero. This improves gradient flow during backpropagation, contributing to a more stable optimisation process and reducing the risk of the 'dying neuron' problem. Within the proposed Res-SE block, the Swish activation is applied after the Batch Normalization layer,

enhancing the non-linear representation capability of feature maps prior to the channel attention (Squeeze-and-Excitation) mechanism. Consequently, the network can preserve weak yet distinctive feature patterns. Furthermore, it can learn richer hierarchical feature representations and achieve more stable convergence throughout the training process.

The final output is obtained using the formula specified in Equation (14). After the residual function and shortcut output are summed, they are passed through the Swish activation function. This structure enables the effective learning of multi-layer representations and the efficient training of deep architectures by combining linear and non-linear components.

LSTM Layer: Long Short-Term Memory (LSTM) is a structure proposed by Hochreiter and Schmidhuber in 1997, developed specifically to address the gradient vanishing/exploding issues experienced by traditional Recurrent Neural Networks (RNNs) when learning long-term dependencies. The most fundamental feature of LSTM is its ability to preserve information obtained from past data for extended periods through gate mechanisms that regulate information flow [78]. In this study, the LSTM layer was not used to directly model raw packet-level time series or continuous traffic streams. Instead, each SCADA traffic sample was represented as an ordered numerical feature vector extracted from network traffic and reshaped into a one-dimensional feature sequence. Therefore, the LSTM component was used to learn sequential and contextual relationships among traffic-derived attributes rather than direct temporal dependencies in packet streams. An LSTM cell contains the following elements (see Figure 3):

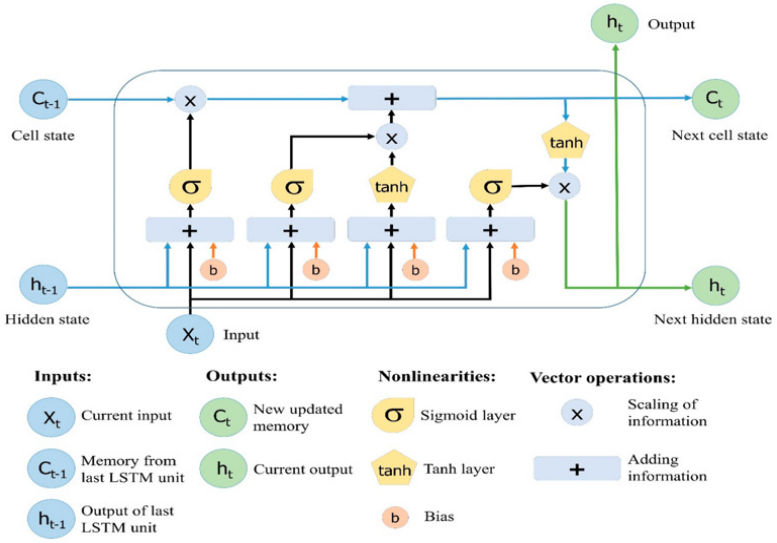


Figure 3. LSTM structure [79].

Cell state (C_t): The main route for transporting long-term information. Hidden state (h_t): Short-term information representing the current output. Forget gate (f_t): Determines which information will be deleted from the memory. Input gate (i_t): Controls which parts of the incoming data are written to memory. Output gate (O_t): Regulates how much of the information in memory is transferred out. Cell candidate ($\widetilde{C}_t$): Creates a candidate for new information that can be added.

The updating process of the LSTM is defined by the following equations:

$f_t = \sigma\left(W_f.[h_{t-1}, x_t] + b_f\right)$: This expression defines the forget gate. It also determines how much of the previous information is retained. $i_t = \sigma(W_i.[h_{t-1}, x_t] + b_i)$: This is the input gate. It regulates the rate at which new information enters the memory. $\widetilde{C}_t = \tanh(W_C.[h_{t-1}, x_t] + b_C)$: This is the candidate memory. It produces new information suitable for addition. $C_t = f_t \odot C_{t-1} + i_t \odot \widetilde{C}_t$: This is the updated cell state,

representing the combination of the appropriate part of the old information and the new information. $O_t = \sigma(W_o.[h_{t-1}, x_t] + b_o)$: This is the output gate. It selects which part of the cell information will be the output. $h_t = o_t \odot \tanh(C_t)$: This is the hidden state, representing the information that forms the final output.

Here, σ denotes the sigmoid function, tanh denotes the hyperbolic tangent function, and $\odot$ denotes element-wise multiplication. This structure enables the LSTM to update information when necessary and store it for extended periods when required. A key factor in the success of LSTM is a mechanism called Constant Error Carousel (CEC), which allows the error signal to be carried over time without loss. This enables the model to access past information even in long sequences. The training process is typically carried out using the Backpropagation Through Time (BPTT) algorithm. Over time, different types of LSTMs have been developed. The basic (vanilla) LSTM consists only of input, forget, and output gates. Peephole LSTM provides additional information transfer by directly connecting the memory state to the gates. Bidirectional LSTM processes data in both forward and backward directions, while Stacked LSTM can learn more complex dependencies by stacking multiple layers on top of each other. LSTM is widely used in many areas such as natural language processing, speech recognition, time series prediction, health data analysis, and network security. Thanks to its ability to learn long-term relationships, it has become an indispensable deep learning model for sequence-based problems [80]. In the proposed architecture, the outputs obtained from the Residual-SE block were fed into an LSTM layer to model feature-level sequential and contextual dependencies within the transformed SCADA traffic representation. During preliminary hyperparameter experiments, different numbers of LSTM units were empirically evaluated. A configuration of 128 units provided the best validation performance while maintaining reasonable model complexity and was therefore selected for the final architecture. The LSTM layer was implemented with return_sequences = True, allowing the model to preserve the sequential feature representation and capture contextual relationships among traffic-derived attributes in addition to the features extracted by the convolutional components.

Pooling and Fully Connected Layers: The output of the LSTM layer was summarized using Global Average Pooling (GAP) and converted into a fixed-length vector. This vector was fed into a dense layer with 128 neurons and ReLU activation, and Dropout (rate = 0.3) was applied to reduce overfitting.

Output Layer: The final layer contains C neurons, where C denotes the total number of target classes in the dataset. In the SCADANet dataset, C = 14, representing one normal traffic class and thirteen attack classes. Each neuron corresponds to one class, and the Softmax activation function converts the output logits into class probabilities. Therefore, the output vector is defined as $p = [p_1, p_2, \ldots, p_C]$, where p_i indicates the predicted probability of the input sample belonging to the i-th class and $\sum_{i=1}^{C} p_i = 1$. The class with the highest probability is selected as the final predicted label.

3.3. Training Strategy

The training of the proposed model within the scope of this study was performed using the Adam optimization algorithm with an initial learning rate of 0.001. Since the dataset includes imbalanced attack classes, the final model was trained using a multi-class focal loss function rather than standard categorical cross-entropy. Focal loss is derived from the cross-entropy formulation and reduces the contribution of easily classified samples while increasing the effect of difficult and minority-class samples during optimization. In this study, the focal loss parameters were set as $\gamma = 2.0$ and $\alpha = 0.25$.

In addition to focal loss, class weights were used to further reduce the negative effect of class imbalance. These weights were calculated based only on the training data of each fold

and were defined in inverse proportion to the number of samples in each class. Therefore, rare attack classes contributed more strongly to the total loss during model training.

During the training phase, several regularization and optimization strategies were used to prevent overfitting, improve generalization, and stabilize the learning process. EarlyStopping was applied by monitoring the validation loss. If no improvement was observed for 15 consecutive epochs, the training process was automatically terminated, and the best model weights were restored. In addition, ReduceLROnPlateau was used to adaptively adjust the learning rate. If the validation loss did not improve for 7 consecutive epochs, the learning rate was reduced by half. Finally, ModelCheckpoint was used to save the model weights corresponding to the lowest validation loss.

The combined use of focal loss, class weighting, early stopping, adaptive learning rate reduction, and checkpointing enabled a more stable and class-balanced training process while reducing the risk of overfitting.

4. Experimental Work

This section presents in detail the experimental evaluations of the proposed model on two different datasets, the cross-validation strategy used, optimizations for the imbalanced data problem, performance metrics, and ablation analyses. The experiments aim to comprehensively demonstrate the generalizability of the method, the contributions of its components, and its robustness under class imbalance.

4.1. Experimental Setup and Acquisition of the SCADANet Dataset

The SCADANet dataset used in this experimental study was generated in a fully isolated virtual SCADA laboratory environment. The dataset obtained at the conclusion of the experimental study consists of a total of 486,714 labeled network packets. The dataset includes packets obtained from 13 different attack scenarios along with normal traffic, containing a total of 14 traffic classes. The dataset contains 6403 distinct source IP addresses, 123 destination IP addresses, and 165 different network protocols. Approximately 83% of the traffic consists of attack packets, while 17% consists of normal traffic packets. Additionally, class imbalance is observed in the dataset. This situation necessitates the use of appropriate training strategies to ensure that underrepresented attack classes are learned correctly.

Although the physical SCADA topology consists of seven virtual machines, the reported diversity of sources and destinations does not correspond to unique physical entities. These values stem from session-level variability caused by temporary source port randomisation, fragmented communication sessions, protocol handshake diversity and variations in JA4+ based behavioural fingerprints. Consequently, the reported traffic diversity reflects logical communication diversity rather than the number of physical nodes. This enables the dataset to preserve realistic communication heterogeneity within a controlled virtualised environment.

In addition to classic network attacks such as TCP SYN Flood, TCP ACK Flood, UDP Flood, ICMP Flood, HTTP Flood, SQL Injection, Port Scan, Vulnerability Scan, and Operating System Scan, the dataset also includes SCADA-specific attacks such as Modbus False Data Injection (FDI), Modbus Command Abuse, APT Data Exfiltration, and Insider Threat. These attacks were designed as part of an experimental study to represent various threat categories encountered in industrial control systems, including reconnaissance activities, resource consumption, service exploitation, vulnerability scanning, protocol manipulation, covert data exfiltration, and insider sabotage.

To reflect real-world industrial network behavior, attacks were embedded within normal traffic flows using various protocol structures such as Modbus/TCP, UDP, ICMP,

and HTTP. As a result, the dataset provides a rich and realistic research environment that encompasses both the normal operational behavior of SCADA systems and multi-class attack patterns. In particular, the inclusion of SCADA protocol manipulation scenarios in the dataset ensures that it represents not only traditional network attacks but also cyber–physical threats targeting industrial communication protocols.

The experimental environment created for this study was configured using VMware Workstation Pro running on the Windows operating system. As shown in Figure 4, this fully isolated virtual laboratory environment is designed as a multi-node SCADA topology consisting of seven virtual machines (VMs). The topology includes attacker systems, target servers, Modbus/TCP-based field devices, SCADA/HMI master control systems, and a monitoring node that passively monitors all network traffic.

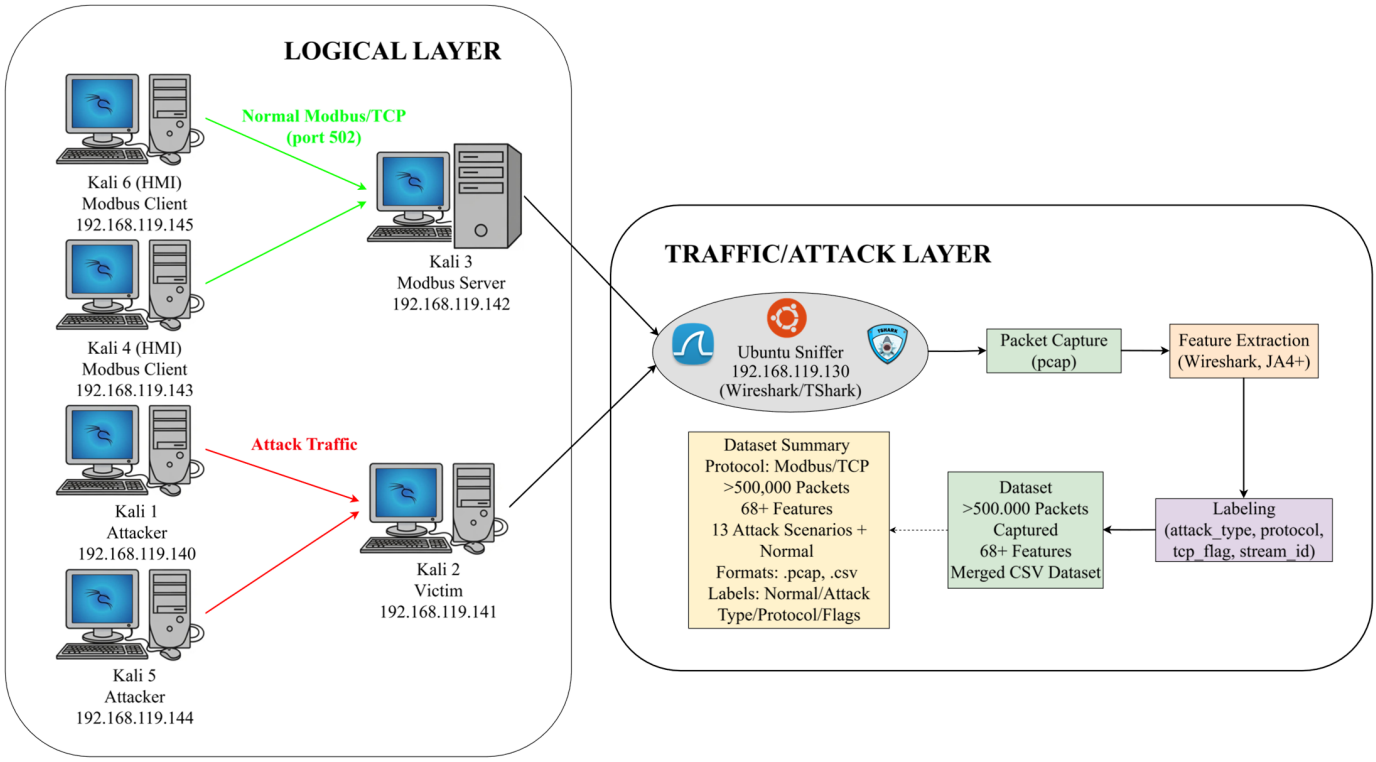


Figure 4. General design of the virtualized SCADA architecture created to obtain the dataset.

All virtual machines were configured to connect to each other via a NAT-based network, forming a single broadcast domain. The Ubuntu-based sniffer virtual machine, configured to capture network traffic, was run in passive mode on the virtual switch and recorded all packets generated within the environment. All network traffic was saved in ".pcapng" format using Wireshark v4.6.3 and TShark v4.6.3 tools, and then multi-layer network features were extracted. The extracted features include source/destination IP addresses, port information, TCP flags, packet lengths, session information, timestamps, and behavioral traffic fingerprint features based on JA4/JA4S/JA4X.

The data labeling process was carried out by synchronizing the timestamps from the attack scripts with the network traffic timestamps. This allowed for the precise determination of attack start and end intervals and reduced the likelihood of mislabeling. Additionally, unnecessary ARP broadcast packets and duplicate records were filtered out during the preprocessing stage.

In the virtual environment created for the experimental study, once all attack scenarios and legitimate SCADA communications had been carried out, the resulting network traffic was recorded in ".pcapng" format using the Wireshark and TShark tools. The traffic

data obtained was synchronised with the timestamps derived from the attack scripts to ensure that the start and end times of each attack could be accurately determined. During the pre-processing stage, the final tagging process was carried out by removing unnecessary ARP broadcast packets and duplicate records. The tagged packet records were then grouped according to the relevant attack categories. Furthermore, the number of packets belonging to each class was calculated. The percentage values presented in Table 5 were obtained by dividing the number of packets in each class by the total number of labelled packets (486,714). Consequently, Table 5 shows the final class distribution of the SCADANet dataset following the completion of the traffic collection, pre-processing, timestamp synchronisation and labelling processes.

Table 5. Class Distribution of the SCADANet Dataset.

Traffic Category	Attack Type	Sample Count	Percentage (%)
Normal Traffic	Legitimate SCADA Communication	82,741	17.00
Flooding Attack	TCP-SYN Flood	58,214	11.96
Flooding Attack	TCP-ACK Flood	41,325	8.49
Flooding Attack	UDP Flood	52,884	10.86
Flooding Attack	ICMP Flood	44,632	9.17
Flooding Attack	HTTP Flood	36,911	7.58
Web Attack	SQL Injection	18,406	3.78
Reconnaissance Attack	Port Scan	29,154	5.99
Reconnaissance Attack	Vulnerability Scan	21,486	4.41
Reconnaissance Attack	OS Scan	17,632	3.62
SCADA-Specific Attack	Modbus False Data Injection	24,870	5.11
SCADA-Specific Attack	Modbus Command Abuse	18,995	3.90
Advanced Persistent Threat	APT Data Exfiltration	22,411	4.60
Insider Threat	Sabotage Activity	17,053	3.50
Total		486,714	100

The SCADANet dataset is designed to preserve the natural traffic imbalance observed in real industrial control systems. While flood-based attacks have higher traffic volumes, reconnaissance and SCADA-specific attacks occur at lower intensities. The class distributions for the dataset are presented in Table 5.

To support the reproducibility of the study, the virtual network architecture, traffic generation processes, attack scenarios, traffic collection mechanisms, and preprocessing steps are described in detail. Additionally, the SCADANet dataset has been made available to researchers via the Kaggle platform. The relevant analysis code is also planned to be shared publicly following the acceptance of the paper.

As shown in Table 5, the SCADANet dataset is designed to preserve the natural traffic irregularities frequently observed in industrial control systems. Flood-based attacks exhibit higher packet density, while reconnaissance and SCADA-specific attacks occur at lower densities within the traffic flow. This approach enhances the realism of the dataset. Additionally, it contributes to the more reliable evaluation of attack detection models under realistic SCADA traffic conditions.

The roles of virtual machines used in the virtual laboratory environment and their task definitions are summarized in Table 6.

4.2. Traffic Generation and Communication Flows

Two primary communication flows have been defined in the simulated environment. The first flow is legitimate industrial communication established between the Modbus/TCP server running on Kali 3 and the Modbus clients on Kali 4 and Kali 6. This structure is configured to mimic the periodic query–response cycles observed in real field environments.

The Modbus clients generated deterministic SCADA traffic by performing holding register, input register, and coil read/write operations at specific time intervals. In the experimental setup, the Modbus polling interval was set between 250 ms and 500 ms to reflect typical industrial HMI-PLC communication cycles. Two independent HMI nodes generated asynchronous read and write requests simultaneously to simulate the multi-operator control and distributed control behaviour frequently observed in real SCADA environments. This configuration maintained deterministic cyclic communication whilst introducing realistic operational variability.

Table 6. Virtual machines used in the experimental virtual laboratory structure created to obtain the dataset.

Virtual Machines	Role	Task Description	IP Address
Ubuntu VM	Listener (Sniffer)	It is configured to passively capture all traffic on the network using Wireshark and Tshark.	192.168.119.130
Kali VM 1	Attacker	They have run attack scripts to simulate malicious activities.	192.168.119.140
Kali VM 2	Victim	It hosted a web server and served as the main target of attacks.	192.168.119.141
Kali VM 3	Modbus Server	It has simulated a SCADA field device by running a Modbus/TCP server.	192.168.119.142
Kali VM 4	Modbus Client	It represented a SCADA/HMI master system that generates legitimate Modbus traffic.	192.168.119.143
Kali VM 5	Attacker	It has run additional attack scripts to create malicious traffic scenarios.	192.168.119.144
Kali VM 6	Modbus Client	It has served as a second SCADA/HMI master system that generates standard Modbus communication.	192.168.119.145

The second flow is attack traffic. Attacker nodes located on Kali 1 and Kali 5 performed various attack scenarios targeting the target system running on Kali 2. All this traffic was generated simultaneously and interleaved without interrupting normal Modbus communication. This approach ensured that realistic traffic patterns were obtained, with attack traffic embedded within normal industrial flow.

During normal traffic generation, not only Modbus/TCP communication but also ICMP-based availability checks and HTTP-based service requests commonly observed in industrial networks are included in the system as background traffic. Thus, the resulting dataset represents a multi-service and heterogeneous SCADA network rather than a synthetic structure based on a single protocol.

4.3. Traffic Capture and Feature Extraction

The network traffic capture process was performed passively using the TShark tool running on an Ubuntu VM. Separate ".pcapng" files were created for each scenario, and Layer-2 noise traffic (such as ARP packets) was filtered during the analysis phase.

The captured raw packets were parsed using Wireshark and TShark; attributes belonging to the frame, IP, TCP/UDP/ICMP, and application layers were extracted in detail. Additionally, cryptographic and behavioral fingerprint characteristics were obtained at the session and flow levels using the JA4+ mechanism.

The SCADANet dataset obtained from the experimental study was designed to include both low-level packet metadata and protocol-aware traffic characteristics in order to represent the heterogeneous structure of SCADA network communications. Frame-level features such as timestamps, protocol identifiers, packet lengths, TCP flags, sequence-related information, and JA4^{+}-based traffic fingerprint features were extracted. Additionally, it includes attributes found at the IP layer, the TCP/UDP/ICMP layer, and the application

layer. JA4$^+$-Based Behavioural Fingerprint Features: Traditional network features represent packet headers and traffic statistics. JA4$^+$ generates behavioural fingerprints by utilising information derived from protocol handshake processes occurring during communication, without the need to examine packet contents. Let $P = \{f_1, f_2, \ldots, f_n\}$ denote the set of protocol-specific features extracted from a network session. The JA4+ fingerprint is obtained as follows, using a hash function applied after protocol normalisation $N(\cdot)$:

$$JA4^+ = H(N(P))$$

In the formula, $H(\cdot)$ denotes the fingerprint generation function. This method produces stable identifiers representing the behavioural characteristics of communication endpoints, independent of the encrypted packet contents. Eight JA4$^+$-based features (JA4H, JA4H_R, JA4S, JA4S_R, JA4T, JA4L, JA4X and JA4SSH), obtained from the SCADANet dataset using the official JA4+ Wireshark plugin, were used in conjunction with traditional network and application layer features. These behavioural fingerprints make a significant contribution to the detection of low-volume and covert attacks, particularly APT-based data exfiltration, insider threats and Modbus command exploitation. This is because, whilst such attacks often exhibit flow statistics similar to normal traffic, their protocol negotiation behaviours and the behavioural fingerprints of communication endpoints display distinct differences. Following preprocessing, the dataset was constructed to include high-dimensional feature representations encompassing both numerical and categorical traffic attributes. To reduce dimensionality and computational load, fields with high cardinality and low information content were removed during the preprocessing stage. However, distinctive traffic features were retained through feature selection and encoding processes. This developed preprocessing strategy is suitable for deep learning-based cyberattack detection. As a result, it enabled the creation of compact and computationally efficient feature representations.

These characteristics provide distinctive information, particularly for detecting low-volume and covert SCADA attacks.

4.4. Labeling Process

The data labeling process is not based solely on port or protocol-based heuristic rules. Instead, the start and end times of attacks were precisely determined by synchronizing the log records of the scripts that initiated the attack scenarios with the packet timestamps. This approach increased the accuracy of class labels and ensured that the dataset can be used reliably in multi-class attack detection studies.

As a result, each packet was labeled as either “normal traffic” or the relevant attack class and structured in CSV format for use in the analysis phase.

4.5. Traffic Generation and Attack Scenario Design

In order to evaluate the resilience of attack detection mechanisms in industrial environments, a cyber-attack scenario was designed to represent both threats targeting traditional information technologies and attacks capable of affecting industrial control infrastructures. The attack modelling process was carried out in a controlled virtual SCADA environment where normal industrial communication traffic and malicious activities were generated simultaneously.

In the experimental study, the traffic generation process in the SCADA system was carried out in two stages. In the first stage, harmless traffic representing normal operating conditions was generated and recorded as reference data. In this stage, continuous Modbus/TCP communication was generated between SCADA client and server nodes to reflect realistic industrial communication behaviour. This communication represents the

control and data exchange between SCADA-based monitoring and control systems and the Programmable Logic Controllers (PLCs) on site. Additionally, ICMP and HTTP traffic was generated in the background to mimic auxiliary services commonly found in industrial network environments.

In the second stage, various cyber-attack scenarios were implemented whilst normal traffic continued to operate in the background. This approach enabled the creation of mixed traffic environments in which malicious activities were embedded within legitimate SCADA communication traffic. Consequently, attack conditions that might arise in real operational environments where industrial processes continue to run were modelled more realistically. All network traffic was passively captured and subsequently recorded for analysis and tagging.

The attack scenarios implemented include TCP-SYN Flood, TCP-ACK Flood, UDP Flood, ICMP Flood, HTTP Flood, SQL Injection, Port Scan, Vulnerability Scan and OS Scan attacks. These attacks were selected to represent different threat categories, such as reconnaissance activities, resource consumption and application-layer exploitation. In particular, Port Scan, Vulnerability Scan and OS Scan attacks represent reconnaissance activities aimed at identifying open services, system configurations and potential security vulnerabilities within industrial network infrastructure. In contrast, TCP-SYN Flood, TCP-ACK Flood, UDP Flood, ICMP Flood and HTTP Flood attacks simulate denial of service conditions aimed at disrupting the availability of critical SCADA services by consuming network bandwidth or system resources. The SQL Injection attack, meanwhile, models application-layer exploitation targeting web-based management interfaces integrated with modern SCADA monitoring systems.

In addition to these traditional IT-based threats, SCADA-specific attack scenarios targeting industrial control systems were also incorporated into the experimental environment. These attacks include Modbus False Data Injection (FDI), Modbus Command Abuse, APT Data Exfiltration and Insider Threat (Sabotage). The Modbus False Data Injection (FDI) attack simulates the manipulation of process data transmitted between industrial devices and SCADA systems, representing situations that could lead to incorrect operational decisions. The Modbus Command Abuse attack, on the other hand, refers to the sending of unauthorised control commands to field devices via the Modbus protocol. The APT Data Exfiltration scenario models the covert and prolonged exfiltration of sensitive operational data outside the industrial network. The Insider Threat (Sabotage) scenario, on the other hand, represents an authorised user engaging in malicious behaviour to disrupt system operations or manipulate control processes.

Each attack scenario was carried out over periods ranging from 5 to 20 min, depending on the type of attack. Reconnaissance attacks were conducted in short bursts at intervals to mimic realistic scanning behaviour, whilst flood-based attacks were run over longer periods to generate sustained abnormal traffic patterns. SCADA-specific attacks, on the other hand, were carried out by directly interacting with industrial communication channels, thereby enabling the observation of abnormal command patterns and data manipulations that could occur in Modbus-based control traffic.

The dataset created as part of the experimental study comprises a total of fourteen traffic categories, consisting of one normal traffic class and thirteen different attack classes. By combining traditional network attacks with SCADA-specific threat scenarios within the same industrial communication environment, the dataset provides a comprehensive testing environment for evaluating machine learning- and deep learning-based intrusion detection systems developed for industrial control networks.

The SCADANet dataset, compiled from the results of current experimental studies, includes covert APT-style data exfiltration and process-focused Modbus manipulation

attacks; future expansions will also incorporate fully coordinated, multi-stage infiltration campaigns encompassing the stages of reconnaissance, lateral movement, persistence and sequential cyber–physical attacks.

4.6. SCADA-Focused Design of the Proposed Architecture

The deep learning-based multi-layer architecture proposed in this study is applicable to general network intrusion detection problems. Furthermore, the experimental environment and traffic modelling approach used in this study have been specifically designed to reflect the operational characteristics of SCADA systems. The dataset used in the study was created by modelling a Modbus/TCP-based SCADA architecture representing the communication between Human–Machine Interface (HMI) systems and field devices in a virtualised environment. This structure reproduces the deterministic query–response communication cycles commonly observed in industrial monitoring and control processes. This communication structure differs significantly from traditional corporate network traffic.

Although Kali Linux was selected as the underlying operating system for the Modbus client and server virtual machines, its purpose was not to emulate proprietary SCADA software platforms or industrial PLC firmware. Instead, legitimate industrial communication was generated using the pyModbusTCP library, which implements the standard Modbus/TCP protocol independently of the host operating system. Consequently, the captured network traces reflect protocol-compliant Modbus/TCP communication rather than operating system-specific behaviour. The use of a virtualised software-defined environment provides a reproducible, flexible, and cost-effective platform for generating controlled industrial network traffic. Nevertheless, vendor-specific implementations and hardware-dependent timing characteristics of commercial SCADA devices remain beyond the scope of the present study.

In addition to normal legitimate operational traffic, the dataset includes both traditional network attacks and threat scenarios specific to SCADA systems. In this context, the model incorporates not only protocol-level attacks such as Modbus False Data Injection and Modbus Command Abuse, but also more complex attack scenarios such as APT-based data exfiltration and insider threats (sabotage). Such attacks directly target industrial communication processes and control commands, representing realistic cyber–physical threats that could arise in critical infrastructure systems.

Furthermore, the experimental environment has been configured to reflect mixed traffic conditions where attack traffic is generated concurrently with normal Modbus communication. This enables the proposed detection approach to learn both normal industrial communication behaviour and the abnormal behaviour resulting from attacks under realistic operational conditions.

Consequently, the contribution of this study is not limited to the multi-component deep learning architecture alone. It is also significant in that it presents an integrated attack detection framework, incorporating a SCADA-focused traffic generation environment and a protocol-aware attack modelling approach. This approach enables the detection of both general network attacks and SCADA-specific anomalies and threats within industrial control networks.

Furthermore, attack detection in SCADA networks is not limited to identifying anomalies at the packet level. In industrial control systems, network traffic is directly linked to the monitoring and control of physical processes. Consequently, attacks often take the form of cyber–physical anomalies that affect not only network behaviour but also the operational control flow. The traffic modelling approach and experimental scenarios used in this study have been designed to preserve this relationship between network communication and control processes. Consequently, the proposed architecture is capable of learning not only

general computer network-based attacks but also behavioural anomalies that could affect industrial control processes. This enables the proposed approach to contribute to the detection of complex attack behaviours that may arise in critical infrastructure environments, such as SCADA systems.

4.7. Second Dataset Used for Comparison (WUSTL-IIoT-2021)

The WUSTL-IIoT-2021 dataset was also used for comparison purposes in this study. This dataset was created in an isolated and virtualized laboratory environment to represent network traffic from industrial IoT-based SCADA systems. In addition to normal operation scenarios, attack scenarios were generated in a controlled manner, and background traffic was preserved.

The dataset contains approximately 1.2 million network flow records and 44 network/flow-based attributes. The records are labeled into five classes: normal, Backdoor, Command Injection, Denial of Service, and Reconnaissance. The explicit definition of Command Injection and Backdoor attacks specific to SCADA systems makes the dataset an important reference for industrial cybersecurity studies. Furthermore, the class imbalance observed in the dataset provides a realistic testing environment for detecting rare attacks and evaluating model robustness.

In this study, two different datasets were used to evaluate the model's performance on both an internally created dataset and an external dataset commonly used in the literature:

- SCADANet Dataset: An original dataset collected within the scope of the study, containing examples from different classes.
- WUSTL-IIoT-2021 [81]: A dataset frequently used in the literature, with high class diversity and a more heterogeneous structure.

For both datasets, all examples underwent standard preprocessing steps (scaling, label encoding, reshaping, etc.) as described in the Methods Section.

To ensure a reliable and statistically meaningful evaluation of the model, the 5-fold cross-validation method was used in all experiments. Accordingly, the dataset was divided into five equal sub-sections while preserving the class distribution; in each experiment cycle, one of these sub-sections was designated as the test set, while the remaining four were used for training. After all layers were completed, the performance values obtained from each fold were combined, and the final results were reported by calculating the arithmetic mean and standard deviation of the metrics. This approach aims to ensure the model's stability, generalizability, and statistical reliability across different data subsets.

The model was trained using the same hyperparameters, the same training strategy, and the same data splitting protocol in all experiments, thus providing a consistent basis for comparison between models.

4.8. Addressing the Imbalanced Data Problem

Due to a significant imbalance between classes in the datasets, a two-stage loss optimization strategy was applied to increase the representativeness of minority classes and improve the model's sensitivity to these classes. In this context, Focal Loss was first used to reduce the impact of easy examples belonging to frequently encountered classes on the loss function, thereby ensuring that the model focused on more difficult examples and minority classes. Additionally, class weights were defined to balance each class's contribution to the training process; these weights were determined in proportion to the inverse of the number of examples in the relevant class, thereby increasing the impact of rare classes on the total loss. This dual-pronged approach reduced biases arising from the imbalanced data problem and significantly strengthened the model's learning capacity, particularly for classes with low sample sizes.

This two-stage approach to the imbalanced data problem has led to significant improvements in both recall and F1-score, especially for classes with small sample sizes.

4.9. Evaluation Metrics

The model's performance was evaluated not only based on accuracy but also using the metrics specified in Equations (15)–(18) to more accurately reflect the effects of class imbalance.

$$Accuracy = (TP + TN)/(TP + TN + FP + FN) \quad (15)$$

$$Precision = TP/(TP + FP) \quad (16)$$

$$Recall = TP/(TP + FN) \quad (17)$$

$$F1 - Score = (2 \times Precision \times Recall)/(Precision + Recall) \quad (18)$$

In these metrics, TP represents true positives; TN represents true negatives; FP represents false positives; and FN represents false negatives. These four fundamental terms form the basis for calculating Accuracy, Precision, Recall, F1-score, and Confusion Matrix. These metrics have been calculated as both class-based and averages (macro/weighted); the mean and standard deviation values of the metrics across folds are reported in tables. Confusion matrices are presented in figures to detail the confusion between classes.

4.10. Results on the SCADANet Dataset

Five-fold cross-validation experiments conducted on the SCADANet dataset obtained from experimental studies demonstrate that the proposed model possesses both high generalization performance and a strong representation capacity that can effectively perform inter-class discrimination. The integrated use of dilated CNN structures enabling multi-scale feature extraction, attention mechanisms at the channel level, and LSTM layers modeling feature-level sequential and contextual relationships within the transformed SCADA traffic representation ensured that the model exhibited stable and consistent performance despite the pronounced class imbalance observed in the dataset.

The model's five-fold average performance metrics on the SCADANet and WUSTL-IIoT-2021 datasets are presented in an integrated form in Table 7.

Table 7. Five-Fold Average Performance Results for the SCADANet and WUSTL-IIoT-2021 Datasets.

Metric	SCADANet	WUSTL-IIoT-2021
Average Accuracy	99.80	99.99
Macro-Precision	98.63	97.79
Macro-Recall	98.70	97.68
Macro-F1-score	98.65	97.70

As shown in Table 7, the proposed model achieved an average accuracy of 99.80%, indicating highly stable and consistent classification performance across the five cross-validation folds. The high macro-averaged metrics demonstrate that the model does not merely focus on majority classes but can also effectively learn patterns from classes with relatively lower sample distributions. In particular, the Macro-Recall value of 98.70% indicates that the model successfully captures a large proportion of samples belonging to minority classes, highlighting its strong sensitivity to rare attack categories. Furthermore, the Macro-Precision value of 98.63% shows that the model maintains a low rate of false positives across different attack classes, suggesting reliable discrimination between normal traffic and multiple attack types. The Macro-F1-score of 98.65%, which balances both

precision and recall, confirms that the model achieves a strong and balanced classification performance across all classes. Overall, these results demonstrate that the proposed architecture can effectively represent the underlying distribution of the SCADANet dataset while maintaining high classification accuracy and balanced performance under multi-class and moderately imbalanced traffic conditions.

The confusion matrix obtained by combining all test results of the five folds is presented in Figure 5. This matrix allows for a detailed examination of the model's confusion levels between classes and clearly shows how errors are distributed across classes, especially in situations where class imbalance is intense.

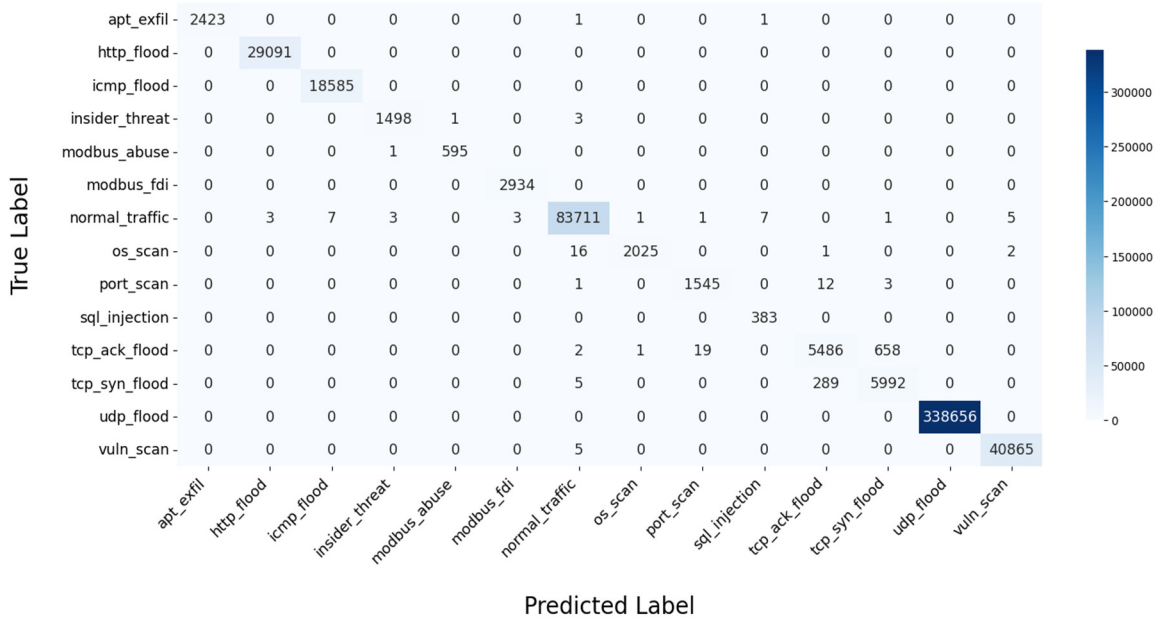


Figure 5. The 5-Fold Combined Confusion Matrix for the SCADANet Dataset.

The 5-fold pooled confusion matrix presented in Figure 5 clearly shows the model's class-based decision behavior and where errors are concentrated among classes. Upon examining the matrix, it is observed that values on the diagonal are dominant, especially in classes with high sample sizes, and off-diagonal errors are very limited; this indicates that the model has strongly learned the distinctive patterns associated with these classes. The low level of false-negative (FN) values in minority classes indicates that the Focal Loss + class weights strategy used to address the imbalanced data problem achieved the desired effect and reduced the model's tendency to miss critical examples. On the other hand, it is observed that class mix-ups are not randomly distributed; rather, they cluster between specific class pairs that can be considered structurally close to each other (e.g., shifts to neighboring classes in some classes). This finding suggests that the model generally learns the decision boundaries clearly, but that patterns with limited overlap between similar classes can cause errors. Overall, Figure 5 confirms that the model performs with high accuracy in majority classes while maintaining sensitivity in minority classes, exhibiting a balanced error profile.

4.11. Results on the WUSTL-IIoT-2021 Dataset

Five-fold cross-validation experiments conducted on the WUSTL-IIoT-2021 dataset demonstrate that the proposed model exhibits high stability across different data distributions and possesses strong generalization capabilities. Although the class distribution of the WUSTL-IIoT-2021 dataset shows a more heterogeneous structure compared to the primary dataset, it was observed that the model's learning capacity and the strategies

applied to address the imbalanced data problem remained effective on this dataset. The model's 5-fold average performance metrics on the WUSTL-IIoT-2021 dataset are presented in Table 7.

The results presented in Table 7 show that the proposed model demonstrates high and balanced performance not only in the majority classes of the dataset but also in classes with lower sample sizes. The fact that the macro-average Precision, Recall, and F1-score values are above 97% reveals that the model has a strong discrimination capacity even in an evaluation environment where all classes contribute equally. In contrast, the fact that all weighted average metrics are at the 99.9973% level shows that the model can represent the distribution of the majority class almost flawlessly and that the overall classification accuracy is exceptionally high.

The confusion matrix obtained by combining the test results of all folds is presented in Figure 6. This matrix allows for a detailed examination of the model's levels of confusion between classes.

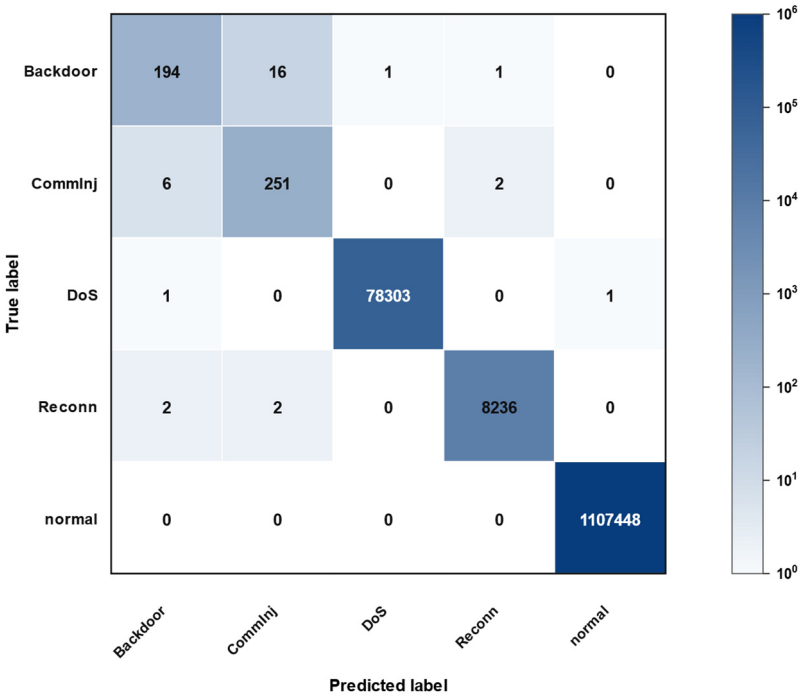


Figure 6. The 5-Fold Combined Confusion Matrix for the WUSTL-IIoT-2021 Dataset.

When examining the confusion matrix presented in Figure 6, it is clear that the model performs with extremely high accuracy on both small and large sample classes. Particularly in the majority class, the fact that nearly all of the over 1.1 million examples are correctly classified demonstrates the model's high representativeness and discriminative capacity. In medium-sized classes, error rates remain negligible, with only a limited number of false positives or false negatives observed. Even in classes with lower sample sizes, false-negative values remain at a minimum level, indicating that the model does not miss critical classes and offers high sensitivity. Overall, the extremely low level of misclassifications confirms that the decision boundaries between classes were learned in a clear and consistent manner.

When these findings are evaluated together, the results obtained on the WUSTL-IIoT-2021 dataset clearly demonstrate that the proposed model offers a high level of generalizability not only to the dataset used in the training process but also to external datasets with different distributions. The high macro-average metrics indicate that minority classes are strongly represented, while the nearly flawless results obtained in the weighted average metrics confirm the model's overall classification stability and reliability. Considering the

low error rates in the confusion matrix, it can be said that the proposed approach offers a scalable, reliable, and stable classification solution for different data sources.

4.12. Ablation Study

Ablation study was performed only on the original SCADANet developed within the scope of this study to quantitatively demonstrate the contribution of the basic components in the proposed architecture to classification performance. In this context, the model architecture was systematically retrained with certain components disabled, and the performance of each configuration was analyzed comparatively. Thus, the contribution of each module to multi-scale spatial representation and feature-level sequential/contextual relationship modeling could be evaluated separately.

The model configurations evaluated in the ablation study are listed below:

- The proposed model including all components (ALL model),
- The model with the Residual and SE blocks removed (Without Res-SE Block),
- The model with the LSTM-based feature-level sequential modeling component removed (Without LSTM),
- The model with the dilated CNN structure, which enables multi-scale feature extraction, removed (Without Dilated CNN). The performance results obtained are presented in Table 8.

Table 8. Ablation Study Results.

Model Configuration	Accuracy	Precision	Recall	F1-Score
ALL model	99.80	98.63	98.70	98.65
Without Res-SE Block	99.74	98.31	98.34	98.30
Without Dilated CNN	99.67	97.75	97.94	97.84
Without LSTM	99.59	97.29	97.38	97.33

The ablation results presented in Table 8 clearly demonstrate that the ALL model configuration, in which all architectural components are active, achieves the highest values in terms of all performance metrics. The model reaches 99.80% accuracy, 98.63% precision, 98.70% recall, and a 98.65% F1-score, confirming the effectiveness of the integrated architecture. This finding shows that the holistic design of the proposed architecture, which considers the interaction between components, provides meaningful and consistent contributions to classification performance.

The findings obtained by disabling different architectural components within the ablation study are summarized as follows:

- When the residual and SE blocks were removed, a decrease was observed in all performance metrics, with the model achieving 99.74% accuracy, 98.31% precision, 98.34% recall, and a 98.30% F1-score. This clearly demonstrates the critical role of attention mechanisms at the channel level in highlighting discriminative features and preserving information flow in deep network architectures. Disabling these components limited the model's discriminative representation capacity, resulting in weaker learning of inter-class decision boundaries.
- Removing the LSTM layer caused the most noticeable decline among the tested configurations, reducing the performance to 99.59% accuracy, 97.29% precision, 97.38% recall, and a 97.33% F1-score. This finding demonstrates that the LSTM structure makes an important contribution to effectively modeling feature-level sequential and contextual relationships within the transformed SCADA traffic representation. Failure to transfer sequential contextual information among traffic-derived attributes to the model led to

an increase in false negatives, particularly by reducing the capture rate of examples belonging to minority classes.

- When the Dilated CNN architecture was removed, the model performance decreased to 99.67% accuracy, 97.75% precision, 97.94% recall, and a 97.84% F1-score. This indicates that the model's ability to represent multi-scale spatial patterns weakened significantly. Without this architecture, which provides a wider receptive field, the model was less capable of distinguishing complex structures occurring at different scales.

In general, the ablation study clearly demonstrates that each architectural component contributes positively to the final model performance. The combination of multi-scale spatial feature extraction (Dilated CNN), channel-wise attention mechanisms (SE blocks), and feature-level sequential/contextual relationship modeling (LSTM) enables the model to achieve high accuracy (99.80%) and strong macro-level classification performance (F1-score: 98.65%) while maintaining balanced detection capability across different attack classes.

Although the obtained overall accuracy values are notably high, the ablation analysis provides additional evidence that the model's performance is not due to trivial class separability or dataset bias. The consistent and systematic performance degradation observed when removing key architectural components indicates that classification success depends heavily on multi-scale spatial modeling, channel-level attention mechanisms, and feature-level sequential/contextual relationship learning. Furthermore, all results are 5-fold cross-validation averages rather than single train–test splits, which reduces the likelihood of favorable partition effects. Additional validation on the WUSTL-IIoT-2021 dataset, which has different traffic characteristics, further supports the idea that the reported performance reflects genuine representation learning rather than artificial separability inherent to controlled experimental settings. Nevertheless, we acknowledge that validating the model in fully operational industrial environments is an important direction for future research.

4.13. Computational Complexity and Deployment Feasibility

The proposed hybrid architecture integrates three parallel dilated one-dimensional convolutional (Conv1D) branches, a Residual block enhanced with Squeeze-and-Excitation (SE) channel attention, and a single-layer Long Short-Term Memory (LSTM) unit. Since industrial cybersecurity deployments may involve hardware constraints, an analysis of computational footprint and deployment feasibility is essential.

Let L denote the input sequence length (i.e., the number of numerical traffic features represented as a one-dimensional sequence). For a Conv1D layer with kernel size K, input channel dimension C_{in}, and output channel dimension C_{out}, the number of trainable parameters is:

$$P_{Conv1D} = K \cdot C_{in} \cdot C_{out} + C_{out}$$

where the second term corresponds to bias parameters. In the implemented model, $C_{in} = 1$, as each traffic instance is modeled as a single-channel feature sequence.

Three parallel dilated Conv1D layers (kernel size K = 3, $C_{out} = 64$) are employed to capture multi-scale spatial patterns. Dilation enlarges the receptive field without increasing parameter count. Their outputs are concatenated and projected back to 64 channels using a 1×1 Conv1D layer, which efficiently reduces dimensionality while preserving representational capacity.

The Residual-SE block contains two Conv1D layers with 64 input and output channels. The embedded SE mechanism performs channel-wise recalibration using a bottleneck structure with reduction ratio r = 8. Although this attention module enhances feature selectivity, it introduces only a limited number of additional parameters relative to convolutional components.

For the LSTM layer, let D denote the input feature dimension (here D = 64) and H the number of hidden units (here H = 128). The total number of LSTM parameters is:

$$P_{LSTM} = 4(DH + H^2 + H) \quad (19)$$

This reflects the four internal gating mechanisms of the LSTM architecture.

Empirical model inspection indicates that the complete architecture contains 156,818 total parameters (156,178 trainable), corresponding to approximately 0.60 MB memory usage under single-precision floating-point representation (FP32) and 0.30 MB under half-precision (FP16). These values demonstrate that the proposed architecture is lightweight compared to many contemporary deep learning models used in network intrusion detection.

From a computational perspective, convolutional layers scale approximately with $O(L \cdot K \cdot C_{in} \cdot C_{out})$, while the LSTM layer scales with $O(L \cdot H^2)$. Given the moderate kernel sizes, limited channel dimensions, and single-layer feature-level sequential/contextual modeling, overall computational demands remain controlled.

In practical deployment scenarios, SCADA intrusion detection mechanisms are typically positioned at supervisory monitoring nodes, industrial edge gateways, or dedicated intrusion detection systems where traffic aggregation and inspection occur. The lightweight parameter scale and modest memory footprint of the proposed architecture support feasibility for such deployment layers. For scenarios requiring further resource reduction, model optimization strategies such as structured pruning, quantization, mixed-precision inference, or knowledge distillation can be applied to further minimize computational overhead.

Overall, the architecture achieves a balanced trade-off between representational capacity and deployment practicality within industrial monitoring infrastructures.

In this context, intrusion detection systems used in real-world SCADA environments must meet low-latency and real-time monitoring requirements due to the deterministic communication cycles between controllers and field devices. A review of the literature indicates that, particularly in situations where rapid inference is necessary to prevent operational disruptions, computationally lightweight and efficient machine learning or deep learning models are more suitable for deployment in industrial environments. Such models can operate effectively on edge gateways or higher-level monitoring nodes. Additionally, the literature highlights that integrating AI-based intrusion detection systems with industrial monitoring platforms or SIEM infrastructures contributes to earlier threat detection by enhancing situational awareness in critical infrastructures such as SCADA. Therefore, the proposed model's low computational load and scalable architecture are well-suited to support real-time network traffic analysis within SCADA monitoring infrastructures. Furthermore, the proposed model has the potential to contribute to making industrial cybersecurity systems more resilient.

4.14. Model Decision Analysis

Explainability is essential in SCADA-based critical infrastructure contexts, where operators require transparent reasoning for trust, incident analysis, and compliance-oriented reporting. Therefore, in addition to predictive performance evaluation, we conducted a SHAP-based post hoc explainability analysis to quantify feature-level contributions to the model's decisions.

Specifically, SHAP (SHapley Additive exPlanations) was used to compute feature attribution scores over the trained model outputs. The global importance profile was obtained by aggregating mean absolute SHAP values across the analyzed samples and classes. The results indicate that the model's predictions are predominantly driven by protocol-level characteristics and packet-level structural features in the network traffic.

Among the most influential variables, Udp_length shows the highest attribution value, followed by frame_len, Tcp_stream, Tcp_src_port, and Time. These features indicate that the model strongly relies on packet size characteristics and session-level TCP behavior when distinguishing between normal and malicious traffic. In addition, transport-layer indicators such as Tcp_len, Tcp_seq, Tcp_window_size, and Tcp_ack also contribute significantly to the model's decision-making process. Furthermore, protocol-specific attributes including Icmp_type, Udp_srcport, and Udp_dstport appear among the important explanatory variables, suggesting that the model captures distinctive patterns associated with different communication protocols. Temporal characteristics such as Frame_time_delta also contribute to the explanation profile, indicating that the model is capable of detecting anomalies related to traffic timing behavior.

These attribution patterns are consistent with attack-induced deviations in industrial network traffic, where malicious activities frequently manifest through abnormal packet-size distributions, unusual transport-layer behaviors, and irregular session dynamics. Overall, the SHAP analysis provides interpretable evidence that the proposed architecture captures domain-relevant traffic characteristics rather than relying on spurious correlations.

4.15. Real-Time Feasibility

Real-time operability is a critical requirement in SCADA-based critical infrastructure systems. Therefore, the architecture proposed in this study was not evaluated solely in terms of classification performance. It was also assessed in terms of computational efficiency and deployment feasibility. The literature review highlights that lightweight deep learning models with low latency and reasonable computational costs are more suitable for industrial environments. In particular, it is stated that hybrid architectures based on CNNs and LSTMs offer advantages over high-parameter Transformer-based models.

The architecture proposed in this study consists of parallel dilated Conv1D layers, Residual-SE blocks, and LSTM components. During the architectural design process, the capacity for multi-scale feature extraction was preserved. Additionally, the goal was to balance computational complexity. To this end, parameter growth was limited by using Batch Normalization, 1×1 Conv1D, and Global Average Pooling layers. The training process was carried out using CUDA-accelerated GPUs.

The results of the experimental study, obtained from the SCADANet and WUSTL-IIoT-2021 datasets, clearly demonstrate that the proposed approach offers high accuracy and balanced classification performance. In particular, the high Macro-Recall and Macro-F1-score values indicate that the model can produce effective results even in low-density attack classes. Additionally, the parallel dilated CNN architecture's ability to model both short- and long-term traffic patterns simultaneously provides a significant advantage for real-time SCADA attack detection.

Furthermore, it is determined that operational factors such as hardware constraints, network latencies, and heterogeneous field devices, which may arise in real industrial environments, could impact system performance. Therefore, testing the proposed approach under various hardware conditions, particularly in edge gateways and real-time SCADA monitoring infrastructures, is considered an important area of research.

5. Discussion

5.1. Comparison with Recent Studies in the Literature

Table 9 lists the most commonly used datasets in the literature for attack detection studies in industrial control systems. The table also highlights recent studies conducted using these datasets and the machine learning approaches applied in these studies. A review of the latest studies in the literature reveals that an accuracy of 87% was reported

using GAN-LSTM on the SWAT dataset, a precision of 86.69% using a GAT-based method on the WADI dataset, and accuracy values of 96.93% and 96.6% on the BATADAL and MSU datasets, respectively. Whilst RF and SVM methods achieved approximately 95% accuracy on the Modbus-based CIC-Modbus dataset, studies conducted on IIoT datasets yielded accuracy values of 98% for Edge-IIoTset and 98.72% for WUSTL-IIoT-2021. However, the majority of these studies have been evaluated on datasets with limited attack types or have focused solely on general network attacks.

Table 9. Performance Comparison of Recent Cyber-Attack Detection Studies on SCADA and Industrial IoT Datasets.

Study/Year	Dataset	Protocol/Traffic Type:	Attack Types	Classes	Result (%)	Algorithm
[82]/2025	SwAT	Network traffic and physical process data from a water treatment plant.	Cyber- and physical attacks, including network attacks and process interruptions.	Replay, DoS, Command Injection	Accuracy 87%	GAN-LSTM
[83]/2025	WADI	Similar to SwaT obtained from water distribution systems.	Cyber–physical attacks.	Normal and Anomaly.	Precision 86.69%	Graph Attention Network based Anomaly Detection (GAT-AD)
[84]/2025	BATADAL	Typically includes water distribution systems.	Simulated cyber–physical attacks.	Data Integrity, Availability-Focused Attacks	Accuracy 96.93%	Ensemble Stacking Machine Learning (ESML)
[85]/2024	MSU	Industrial control system data, Modbus/TCP	Various cyber–physical attacks.	Normal and Attack Scenarios.	Accuracy 96.6%	CNN-based Ensemble Learning
[86]/2025	CIC Modbus	Modbus protocol traffic.	Network-based attacks on Modbus communications.	Fuzzing, DoS, Replay, Modbus Command Injection	Accuracy 95%	Random Forest (RF) and Support Vector Machine (SVM)
[87]/2025	Edge IIOTset	Industrial IoT data.	IoT-specific cyber-attacks.	DoS, Data Injection	Accuracy 98%	Transformer + FedNova
[88]/2025	WUSTL-IIoT-2021	Network traffic from Industrial IoT devices.	Multiple cyber-attacks targeting IIoT communication.	DoS, DDoS, Reconnaissance, Injection	Accuracy 98.72%	Factorization Machine + DNN(DeepFM)
Our Study	SCADANet	Network traffic generated from a virtual SCADA environment using the Modbus/TCP protocol.	Cyber- and SCADA-specific attacks targeting network and control processes.	TCP-SYN Flood, TCP-ACK Flood, UDP Flood, ICMP Flood, HTTP Flood, SQL Injection, Port Scan, Vulnerability Scan, OS Scan, Modbus False Data Injection, Modbus Command Abuse, APT Data Exfiltration, Insider Threat.	Accuracy 99.80%	Multi-Component Deep Learning Approach

The SCADANet dataset used in this experimental study offers a more comprehensive structure designed to address these limitations in the literature. The dataset was created within an isolated virtual SCADA environment comprising attacker systems, Modbus servers, SCADA/HMI clients and a listener node monitoring network traffic. Thanks to this proposed architecture, Modbus client–server communication continues in a manner similar to real industrial environments, whilst attacks are carried out within the same network traffic. Consequently, attack traffic was obtained within a realistic traffic flow rather than a synthetic structure isolated from normal industrial communication.

Another key feature of the SCADANet dataset is the diversity of attacks. In addition to common network attacks such as TCP SYN Flood, UDP Flood, ICMP Flood, HTTP Flood, SQL Injection and Port Scan, the dataset also includes threats specific to SCADA systems, such as Modbus False Data Injection, Modbus Command Abuse, APT Data Exfiltration

and Insider Threat. This enables the dataset to represent attacks occurring at both the network level and the control protocol level. In this respect, the SCADANet dataset also distinguishes itself from many datasets currently in widespread use in the literature.

Furthermore, detailed packet features derived from multi-layer network protocols were utilised in the dataset, and high-accuracy labelling was achieved by matching packet timestamps with the logs of attack scripts. This approach enhances the reliability of the dataset and provides a data structure suitable for multi-class attack detection studies.

Many studies in the literature rely on a single machine learning or deep learning model. In contrast, the Multi-Component Deep Learning Approach proposed in this study is designed as a multi-component architecture capable of jointly learning different feature representations of network traffic. The proposed model enables the simultaneous analysis of both temporal and behavioural patterns in network traffic by facilitating the collaborative operation of multiple deep learning components. This approach offers a significant advantage, particularly in distinguishing between attack types with different characteristics. Considering the results in the range of 86–98% shown in Table 9, the performance achieved by the proposed model on the SCADANet dataset represents a significant result, given that it was evaluated on a dataset containing a greater variety of attack types and a more complex traffic structure.

In conclusion, the SCADANet dataset's high diversity of attacks, realistic network traffic structure and detailed labelling approach provide a more comprehensive testing environment compared to datasets in the literature. The multi-component deep learning model developed using this dataset makes a significant contribution to the literature in terms of detecting different types of cyber-attacks in SCADA systems used in critical infrastructure.

It should be borne in mind that the studies summarised in Table 9 were conducted using different datasets, attack scenarios, class distributions, network traffic characteristics and experimental evaluation protocols. Therefore, the reported performance values are provided solely to offer a general overview of current studies in the literature. Furthermore, they should not be interpreted as a direct benchmark comparison. A scientifically more reliable comparison would require all competing methods to be re-run on the same dataset and under the same experimental conditions. This comprehensive comparison is considered an important research direction for our future work.

Figure 7 was prepared to visually support the comparison results presented in Table 9 and to facilitate their interpretation. Figure 7 visually presents a comparison of the proposed method's accuracy rates with current SCADA/IIoT attack detection studies. This graph enables a clearer assessment of the performance differences between the methods whilst more clearly illustrating the position of the proposed model amongst existing approaches in the literature.

As shown in Figure 7, the proposed method achieved the highest classification accuracy (99.80%) among the studies examined. Although these studies were evaluated using different datasets and experimental conditions, the graphical comparison provides a concise visual summary of the state-of-the-art technology. It also demonstrates the competitive performance of the proposed approach.

5.2. Comparison with State-of-the-Art Methods on the WUSTL-IIoT-2021 Dataset

In order to evaluate the effectiveness of the proposed approach within the scope of this experimental study by directly comparing it with methods found in the existing literature, recent studies utilising the WUSTL-IIoT-2021 dataset were examined. WUSTL-IIoT-2021 is a dataset widely used to evaluate the performance of SCADA-based intrusion detection systems and is regarded as a benchmark in the literature. Consequently, comparing the results reported on this dataset enables an objective assessment of the proposed method's

standing against state-of-the-art approaches. In this context, the methods employed and the accuracy values obtained by representative studies in the literature that utilise the WUSTL-IIoT-2021 dataset were examined and compared with the proposed method. The results of the comparative analysis are presented in Table 10.

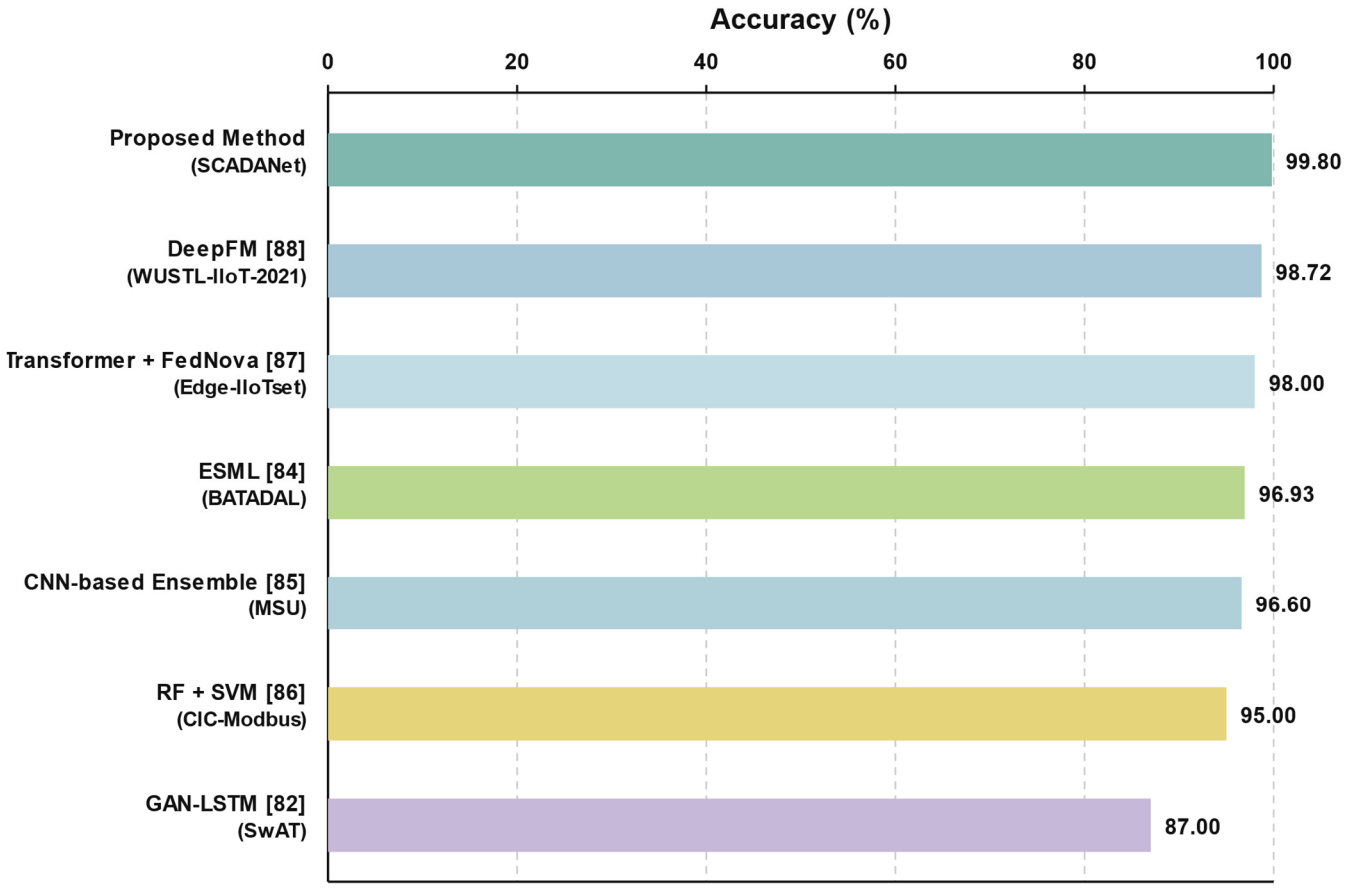


Figure 7. Visual comparison of the proposed model with recent SCADA/IIoT intrusion detection studies in terms of classification accuracy (Siddique et al., 2025 [82]; Lachure and Doriya, 2025 [84]; Naeem et al., 2025 [85]; Vasilica et al., 2025 [86]; Bilal et al., 2026 [87]; Zakariah et al., 2025 [88]).

Table 10. Comparison of the Proposed Method with Existing Studies on the WUSTL-IIoT-2021 Dataset.

Study	Method	Accuracy (%)	Public Dataset	New Dataset	Ablation	Multi-Scale	Attention	Temporal Modeling
[88]	DeepFM	98.72	✓	-	-	-	-	-
[89]	CNN + Transformer	98.75	✓	-	-	✓	✓	-
[90]	Transformer + GAN + AE	97.86	✓	-	-	✓	-	-
[91]	AOA-SMA + Enhanced GRU + Attention	97.90	✓	-	-	-	✓	✓
Proposed	Parallel Dilated CNN + Residual-SE + LSTM	99.99	✓	✓	✓	✓	✓	✓

Note: ✓indicates the presence of the corresponding feature, whereas "–" indicates its absence.

An examination of Table 10 reveals that many studies using the WUSTL-IIoT-2021 dataset have achieved high accuracy rates. However, the majority of these studies have been validated solely on the WUSTL-IIoT-2021 dataset and do not provide a generalisability analysis across different SCADA datasets. Furthermore, most studies do not include an ablation analysis evaluating the components of the proposed architecture.

In contrast, the multi-layer architecture proposed in this study has been evaluated on both the WUSTL-IIoT-2021 dataset and the SCADANet dataset, which was developed to represent real SCADA network traffic. The results demonstrate that the proposed approach exhibits consistent performance across different datasets and provides high generalisability. Furthermore, the ablation analysis conducted confirms the contribution of each component of the proposed multi-layer architecture to performance. In these respects, the proposed approach not only offers high accuracy but also presents a reliable and comprehensive attack detection framework suitable for use in real SCADA environments.

5.3. Comparison of Machine Learning and Deep Learning Methods on the Datasets

Additional experiments were conducted to assess the effectiveness of the proposed architecture against commonly used classification approaches. These experiments used the following models: k-nearest neighbors (KNNs), linear support vector machine (Linear SVM), logistic regression, Gaussian naive Bayes (Gaussian NB), multi-layer perceptron (MLP), one-dimensional convolutional neural network (CNN1D), and standard long short-term memory (LSTM). These methods represent conventional machine learning and deep learning approaches that are commonly used in network intrusion detection.

All models were evaluated using five-fold stratified cross-validation on both the SCADANet and WUSTL-IIoT-2021 datasets. The same input features, fold partitions, preprocessing procedures, and evaluation metrics were applied to each dataset. Missing-value imputation and feature standardization were performed independently within each fold using only the corresponding training data.

Since both datasets have imbalanced class distributions, the primary comparison metrics were macro-precision, macro-recall, and macro-F1-score. These metrics assign equal importance to each class, providing a balanced evaluation of model performance across majority and minority traffic categories. The five-fold mean results are presented in Table 11.

Table 11. Performance Comparison (%) Across Both Datasets.

Model	SCADANet			WUSTL-IIoT-2021		
	Macro-Precision	Macro-Recall	Macro-F1-Score	Macro-Precision	Macro-Recall	Macro-F1-Score
KNN	92.61	92.04	92.32	97.03	96.49	96.74
Linear SVM	69.63	82.94	74.32	96.29	96.63	96.43
Logistic Regression	69.38	90.10	75.03	94.81	97.20	95.90
Gaussian NB	79.86	83.31	78.31	97.00	96.22	96.30
MLP	88.40	93.42	89.30	94.05	96.42	95.11
CNN1D	79.16	86.27	79.66	89.31	89.28	87.54
LSTM	92.58	93.26	92.43	87.51	94.66	88.69
Proposed model	98.63	98.70	98.65	97.79	97.68	97.70

Among the comparison methods, the standard LSTM achieved the highest macro-F1-score on the SCADANet dataset, reaching 92.43%. KNN followed closely behind with 92.32%. Among the standard classifiers, KNN obtained the highest macro-precision, while MLP achieved the highest macro-recall. The proposed architecture achieved macro-precision and recall of 98.63% and 98.70%, respectively, as well as an F1-score of 98.65%. This score was 6.22 percentage points higher than that of the standard LSTM, the strongest comparison method on SCADANet. These results suggest that combining multi-scale convolutional feature extraction, channel-level attention, and contextual modeling provides a more effective representation of heterogeneous traffic patterns in the dataset.

Among the comparison methods, KNN achieved the highest macro-F1-score on the WUSTL-IIoT-2021 dataset, followed by Linear SVM and Gaussian NB. Logistic regression

obtained the highest standard macro-recall, while KNN provided the best balance between macro-precision and recall. The proposed model achieved macro-precision, recall, and F1-score of 97.79%, 97.68%, and 97.70%, respectively, on the WUSTL-IIoT-2021 dataset. Compared to KNN, the best standard classifier on this dataset, the proposed architecture improved the macro-F1-score by 0.96 percentage points.

Overall, the proposed architecture demonstrated the best macro-level performance on both datasets. The performance advantage was more pronounced on SCADANet. Meanwhile, the WUSTL-IIoT-2021 results showed that the model maintained strong, balanced classification performance despite a different feature distribution and class structure.

6. Conclusions and Future Work

- *Proposed Architecture and Dataset:* This experimental study aims to detect cyberattacks targeting SCADA-based critical infrastructure networks. To this end, a multi-layer hybrid deep learning architecture that integrates parallel expanded CNN layers, Residual-SE blocks, and LSTM components has been proposed to effectively detect cyberattacks. The proposed architecture is designed to simultaneously model multi-scale spatial patterns and feature-level sequential/contextual relationships within transformed SCADA traffic representations. Additionally, the SCADANet dataset has been introduced to the literature within a fully isolated virtual SCADA environment where legitimate industrial communications and multi-class attack scenarios are generated concurrently.
- *Experimental Performance:* The proposed multi-layer deep learning architecture was evaluated on both the developed SCADANet dataset and the well-established WUSTL-IIoT-2021 dataset in the literature. Experimental results demonstrate that the proposed deep learning architecture achieves high accuracy, macro-F1-score, and recall values on both datasets. In particular, the high performance achieved in attack classes with few examples demonstrates that focal loss and class weighting strategies enhance the model's sensitivity under imbalanced data conditions. Additionally, the ablation study conducted further demonstrates that each component of the proposed hybrid deep learning architecture makes a significant contribution to the overall classification performance.
- *Key Findings:* The findings obtained in this study demonstrate that the combined use of multi-scale convolutional feature extraction, channel-level attention mechanisms, and sequential modeling components at the feature level enables the effective representation of complex SCADA traffic behavior. Evaluations conducted on different datasets clearly demonstrate that the proposed model is not dependent on a specific data distribution. Experimental results show that the developed deep learning architecture delivers consistent attack detection performance across various industrial traffic structures.
- *Limitations:* However, the study has some limitations. The SCADANet dataset was created in a controlled virtual SCADA environment to ensure experimental reproducibility and systematic evaluation. However, such controlled environments cannot fully reflect the operational complexity of real industrial systems. In particular, hardware diversity, process-level variability, legacy device behaviors, and dynamic network conditions remain limited in this context. The dataset covers a wide range of attacks. However, multi-stage coordinated attacks, long-term covert threats, and advanced cyber–physical manipulation scenarios are not sufficiently represented.
- *Future Work:* Future academic research aims to enhance the realism of the experimental environment. More complex attack scenarios will be incorporated into the dataset. Additionally, the proposed multi-layer deep learning architecture will be validated

in semi-operational and real-world industrial environments. Furthermore, research is planned on low-latency and scalable edge gateway-based deployment strategies. These planned developments are expected to enhance resilience against advanced cyber–physical threats. It is also anticipated that the planned updates will contribute to the development of deployable and reliable AI-based attack detection mechanisms for critical infrastructure systems.

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Data Availability Statement: The WUSTL-IIoT-2021 dataset used in this study is an open-access dataset for research purposes. The dataset can be accessed via the official Washington University in St. Louis (WUSTL) data repository at the following link: https://www.cse.wustl.edu/~jain/iiot2/index.html (accessed on 20 January 2026). In addition to the WUSTL-IIoT-2021 dataset, this study also utilizes the SCADANet dataset, which was developed by the authors to further support the experimental analysis and evaluate SCADA-based intrusion detection scenarios. The SCADANet dataset is publicly available at: https://www.kaggle.com/datasets/ealgul/scada-dataset-v01 (accessed on 20 January 2026). All data analyzed in this study are included in the published article or are publicly accessible through the corresponding dataset repositories.

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